UNIT I

Introduction to Composite Materials & FRP Processing

Definition | classification | applications | lay-up, bag moulding, filament winding, pultrusion, moulding

I. Introduction & Definition

A **composite material** is a material made by combining two or more distinct constituents (different in form or chemical composition, and insoluble in each other) on a **macroscopic scale** so that the combination has properties that are better than those of the individual constituents acting alone.

Examples: wood (cellulose fibres in lignin), bone, reinforced concrete (steel in concrete), glass-fibre/epoxy, carbon/epoxy, Al-SiC.

Why study composites ?

> Metals are isotropic and heavy; modern aircraft, space and sports applications need high **strength-to-weight** and **stiffness-to-weight**.

> Composites let the designer **tailor** the material: choose fibre, matrix, volume fraction and fibre direction to suit the load.

> Properties are *not* fixed: they depend on the design ($E_c \neq E_f, E_m$) - material and structure are made together.

I.1 Constituents of a composite

- **Reinforcement (dispersed phase)** - fibres, particles or flakes. Carries the major load, gives strength and stiffness.
- **Matrix (continuous phase)** - polymer, metal or ceramic. Binds the fibres, transfers load between them, protects them from environment and keeps them in position.
- **Interface** - the bond region between fibre and matrix; governs load transfer, toughness and moisture resistance.

Composite = Reinforcement + Matrix + Interface

I.2 Functions of the matrix

- Holds the fibres together and gives the component its shape.
- Transfers the applied load to the fibres through shear at the interface.

- Separates the fibres, so a crack in one fibre does not run into the next (crack arrest).
- Protects fibres from abrasion, moisture, heat and chemicals.
- Controls inter-laminar shear strength, transverse strength and the maximum service temperature.

1.3 Functions of the reinforcement

- Provides the strength and stiffness: in a good composite fibres carry 70-90 % of the load.
- Controls thermal expansion, thermal and electrical conductivity.
- Gives wear and abrasion resistance (particles, whiskers).

II. Characteristics of Composite Materials

Advantages	Limitations
High specific strength (σ/ρ) and specific modulus (E/ρ)	High raw-material and fabrication cost
Light weight: 20-50 % mass saving in aircraft	Properties vary with batch / process (scatter)
Excellent fatigue and corrosion resistance	Brittle, low impact and transverse strength
Tailorable (anisotropic) properties, directional design	Difficult to repair and to recycle
Low thermal expansion, dimensional stability	Moisture absorption and UV / heat degradation
Part integration, complex shapes in one moulding	Complex analysis; delamination, hard to inspect

2.1 Anisotropy and heterogeneity

Composites are **heterogeneous** (properties change from point to point - fibre or matrix) and usually **anisotropic** (properties depend on direction). A unidirectional lamina is very strong along the fibres (direction 1) but weak across them (direction 2).

E_1 (along fibre) $\gg E_2$ (transverse)

e.g. carbon/epoxy: $E_1 \approx 140$ GPa, $E_2 \approx 10$ GPa

Specific properties (comparison)

> Steel: $\rho = 7.8$ g/cc, $E = 210$ GPa → $E/\rho \approx 27$
> Aluminium: $\rho = 2.7$ g/cc, $E = 70$ GPa → $E/\rho \approx 26$
> Carbon/epoxy: $\rho = 1.6$ g/cc, $E_1 = 140$ GPa → $E/\rho \approx 88$ (3 times better)

III. Classification of Composite Materials

Composites are classified (a) by **matrix**: PMC, MMC, CMC, carbon-carbon; and (b) by **reinforcement shape**: fibrous, laminated and particulate.

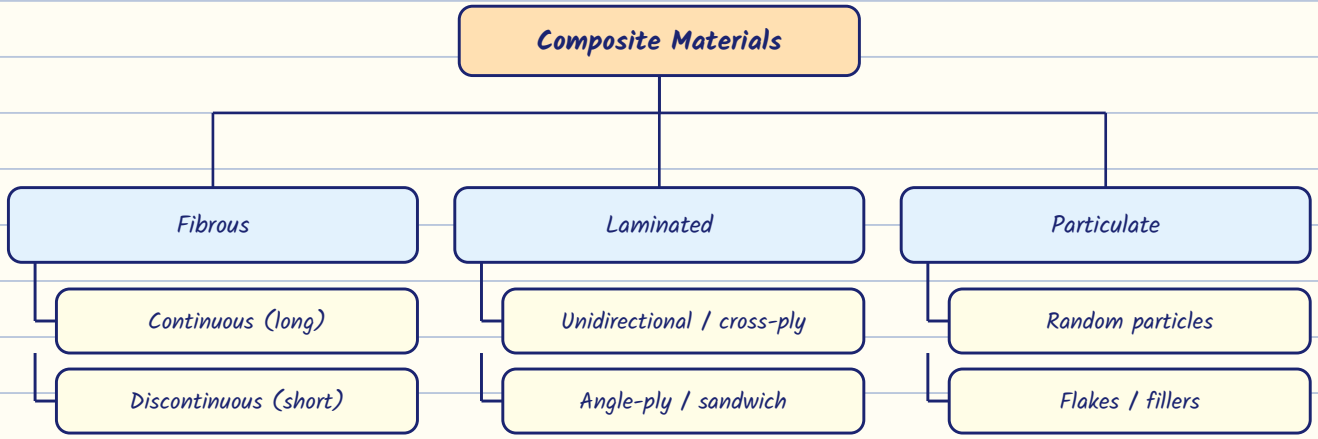


Fig. 1 Classification of composites based on reinforcement

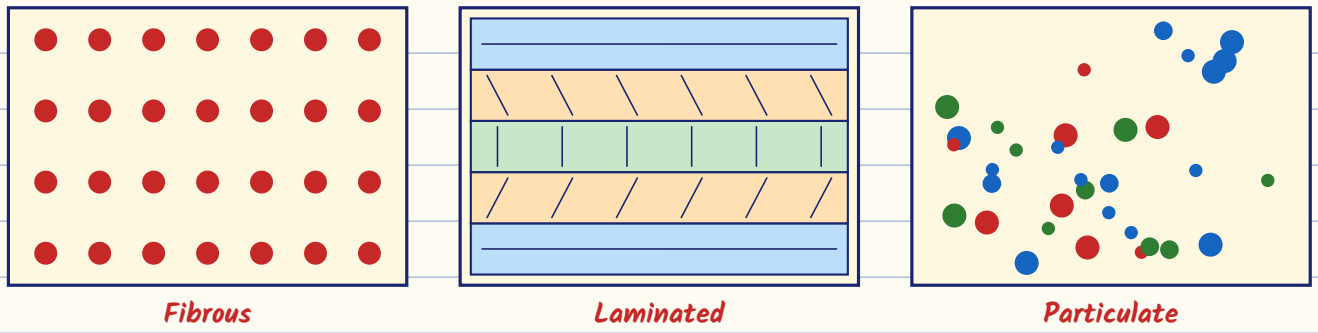


Fig. 2 Sketch of fibrous, laminated and particulate composites

3.1 Fibrous composites

Fibres (length ≫ diameter) in a matrix. **Continuous-fibre** composites (unidirectional, woven, braided) give the highest strength; **discontinuous (short, chopped)** fibres can be aligned or random - lower strength but easy to mould. Strength depends on fibre length, orientation and volume fraction V_f.

3.2 Laminated composites

Made of **layers (plies / laminae)** bonded together, each lamina having its own fibre orientation (0°, ±45°, 90° ...). The stacking sequence is chosen to get the required stiffness in several directions. Also includes **sandwich panels** (thin stiff face sheets + light core of honeycomb or foam) and clad metals, bimetals, plywood.

3.3 Particulate composites

Particles (spherical, flake, irregular) of one material in a matrix of another. Properties are roughly **isotropic**. Used to improve wear, hardness, heat resistance and stiffness: e.g. concrete (sand/gravel in cement), cermets (WC-Co cutting tools), Al-SiC pistons, rubber with carbon black, filler-loaded plastics ($CaCO_3$, talc).

Type	Reinforcement	Main feature	Example
Fibrous	Long / short fibres	Strong in fibre direction	GFRP, CFRP
Laminated	Stacked plies / layers	Tailored in-plane stiffness	Plywood, CFRP laminate
Particulate	Particles, flakes	Isotropic, wear / hardness	Concrete, Al-SiC

IV. Reinforcements and Matrices

4.1 Common fibres

Fibre	Density (g/cc)	Tensile strength (GPa)	Modulus (GPa)	Remarks
E-glass	2.55	2.4-3.4	72	Cheap, most used, good insulator
S-glass	2.50	4.5	86	Higher strength, aerospace
Carbon (HS / HM)	1.75-1.95	3.5-5.5 / 2.5	230 / 400-600	Stiff, light; costly, galvanic with Al
Aramid (Kevlar-49)	1.45	3.6	130	Tough, impact / ballistic; poor in compression
Boron	2.6	3.5	400	Very stiff, big diameter, very costly
SiC / Alumina	2.5-3.9	2-3	300-400	High-temp., MMC / CMC

4.2 Matrix materials

Property	Thermoset (epoxy, polyester, vinylester, phenolic)	Thermoplastic (PEEK, nylon, PP, PPS)
Curing	Irreversible chemical cross-linking; cannot be re-melted	No chemical change; softens on heating, re-formable
Viscosity	Low: easy fibre wet-out	High: difficult impregnation
Properties	Good creep, high temp. stability	Tough, good impact, recyclable
Processing	Long cure cycle (hours)	Fast cycle (minutes), high temperature

Remember: Polyester is cheapest (general GFRP); **epoxy** is the standard aerospace matrix (better adhesion, low shrinkage, 120-180 °C service).

V. Applications of Composites

5.1 Automobile

- Body panels, bumpers, bonnets, doors (SMC / GFRP) - light weight, no rusting, easy complex shapes.
- Drive shafts (CFRP/GFRP), leaf springs, wheels, connecting rods, brake pads, clutch plates.
- F1 monocoque and crash structures (carbon/epoxy); electric-vehicle battery casings. 10 % weight saving → 6-8 % fuel saving.

5.2 Aircraft

- Fuselage, wings, tail, rudder, elevator, floor beams and fairings: composites ≈ 50 % of airframe weight in modern airliners (B787, A350).
- Helicopter rotor blades, propellers, landing-gear doors, radomes, interior panels.
- Military aircraft: carbon/epoxy skins, radar-absorbing (stealth) structures.

5.3 Missiles

- Rocket motor cases (filament wound Kevlar / carbon-epoxy), nose cones, radomes.
- Ablative heat shields (carbon-phenolic), nozzles and launch tubes - high strength/weight, heat resistance, transparency to radar.

5.4 Space hardware

- Satellite structures, antenna reflectors, solar-panel substrates, truss booms - dimensional stability (near-zero CTE of carbon/epoxy).
- Space-shuttle tiles & nose cap (carbon-carbon), payload bay doors, launch-vehicle fairings, pressure vessels.

5.5 Electrical and electronics

- Printed circuit boards (glass/epoxy FR-4), insulators, switch-gear panels, transformer housings, cable trays.
- Electromagnetic shielding (metal-coated fibres), heat sinks (Al-SiC, Cu-C), antenna radomes, wind-turbine blades.

5.6 Marine

- Boat hulls, decks, masts, yachts, canoes, offshore platforms and pipes - corrosion resistance, no rotting, low maintenance.
- Mine-sweepers (non-magnetic), submarine hulls and sonar domes, propellers, storage tanks.

5.7 Recreational and sports equipment

- Tennis and badminton rackets, golf shafts, hockey sticks, fishing rods, archery bows, bicycle frames.
- Skis, snowboards, vaulting poles, cricket bats, helmets, prosthetic limbs: light, stiff and damping.

5.8 Future potential of composites

- **Nano-composites** (CNT, graphene) - very high strength, conductivity; **bio-composites** (natural fibres: jute, flax, hemp) - green and recyclable.
- **Smart / multifunctional** composites with embedded sensors and actuators, self-healing resins; **thermoplastic** composites for fast automated production and recycling.
- Growth in hydrogen tanks, wind & renewable energy, civil infrastructure (bridge decks, rebars), 3D-printed and automated fibre placement for low cost.

VI. Fibre Reinforced Plastic (FRP) Processing

Processing converts fibres + resin into a finished part. The basic steps in every method are: **(1) fibre placement → (2) resin impregnation (wet-out) → (3) shaping on a mould → (4) curing / consolidation → (5) demoulding and finishing.**



Fig. 3 General sequence of FRP processing

6.1 Lay-up and curing

- **Lay-up**: reinforcement (mat, woven roving, fabric, prepreg) is placed ply by ply on the mould in the specified orientation; resin is applied and worked in to remove air.
- **Curing**: the resin changes from liquid to solid by cross-linking, an **exothermic** reaction started by a catalyst / hardener, heat or UV. Cure may be at room temperature (hours/days) or at elevated temperature (80-180 °C) in an oven or autoclave.

- Cure variables: temperature, time, pressure. **Post-cure** improves cross-linking and glass-transition temperature T_g.

6.2 Fabricating process: open mould and closed mould

Open-mould process	Closed-mould process
One mould surface; only one side finished (smooth)	Two mould halves (or mould + bag); both sides finished
Low tooling cost, large parts (boats, tanks)	High tooling cost, high volume small / medium parts
Hand lay-up, spray-up, (wet lay-up, bag-moulding)	Compression, injection, RTM, pultrusion, resin infusion
Slow, labour intensive, emission of styrene	Fast cycle, good dimensional control, low emission
Lower fibre volume fraction (30-40 %)	Higher fibre volume fraction (up to 60-70 %)

6.3 Hand lay-up technique

The oldest, simplest and cheapest method; used for boat hulls, tanks, roofing sheets, bath tubs.

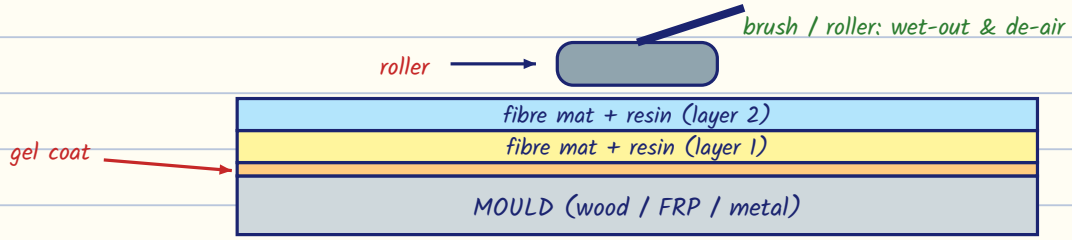


Fig. 4 Hand lay-up

- **Step 1** - clean the mould and apply **release agent** (wax / PVA).
- **Step 2** - spray or brush the **gel coat** (pigmented resin, 0.3-0.5 mm) for a smooth, weather-proof surface; let it gel.
- **Step 3** - place the first glass **mat / woven roving**, pour resin (+ catalyst, accelerator) and **wet it out** with brush or roller.
- **Step 4** - consolidate with a serrated **roller** to squeeze out air bubbles; add layers until thickness is obtained.
- **Step 5** - cure at room temperature (or low heat), demould, trim, finish.

Advantages of hand lay-up	Disadvantages
Low cost tooling, simple equipment	Quality depends on operator skill

Advantages of hand lay-up	Disadvantages
Large and complex shapes, any size	Only one smooth face; slow, labour intensive
Wide choice of fibres and resins	Low V_p porosity, resin-rich areas, health hazard

6.4 Spray-up

Chopped fibre (25-50 mm) and catalysed resin are sprayed together by a **chopper gun** onto the mould, then rolled out. Faster than hand lay-up, fibre is random (short), lower strength - used for bath tubs, panels, containers.

VII. Structural Laminate Bag Moulding

To get higher fibre content and fewer voids than hand lay-up, the laid-up laminate is covered with a flexible **bag** and **pressure** is applied (by vacuum, air or steam) during cure. Used for aircraft skins, radomes, sports goods.

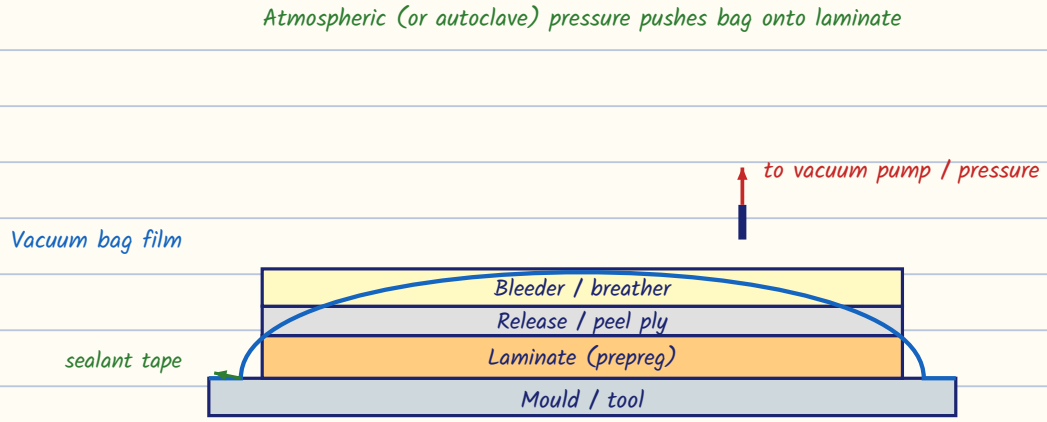


Fig. 5 Vacuum bag moulding (cross-section)

7.1 Types of bag moulding

Process	How pressure is applied	Typical pressure
Vacuum bag	Air evacuated from under bag; atmosphere presses it	≈ 0.1 MPa (1 atm)
Pressure bag	Compressed air / steam inside a bag over the lay-up	0.2-0.4 MPa
Autoclave	Vacuum bag + heated pressure vessel (prepreg)	0.5-0.7 MPa, 120-180 °C

7.2 Production procedure for bag moulding

- 1. Prepare mould surface, apply release agent / gel coat.
- 2. Lay up prepreg or wet plies in the correct stacking sequence and orientation.

- 3. Cover with **peel ply** (release film) → **bleeder / breather** cloth (absorbs excess resin, allows air path).
- 4. Place the **vacuum bag** film, seal edges with sealant tape, connect the vacuum port.
- 5. Apply vacuum (and heat / pressure); **debulk** and cure as per cycle (ramp - dwell - cool).
- 6. Release pressure, remove bag, bleeder and peel ply, demould, trim and inspect.

Exam tip: Learn the bag stack (mould → release → laminate → peel ply → bleeder → breather → bag) and the 3 types of bag moulding with sketches.

VIII. Filament Winding

A continuous process in which **resin-impregnated fibre (roving / tow)** is wound under tension over a rotating **mandrel** while a carriage traverses along the axis. Used for hollow, axisymmetric parts: pipes, pressure vessels, gas cylinders, rocket motor cases, drive shafts.

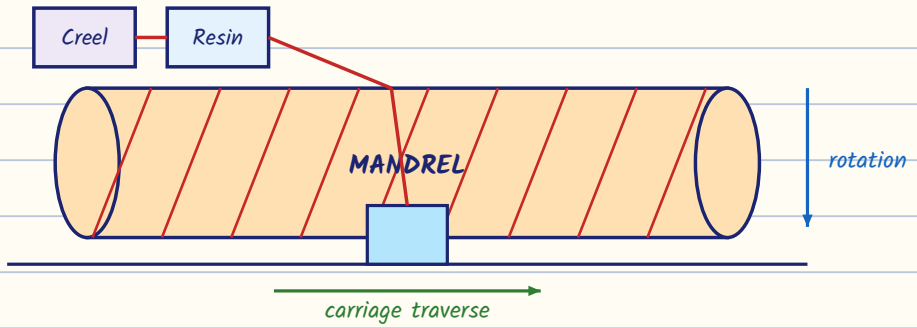


Fig. 6 Filament winding set-up

8.1 Winding patterns

- **Hoop (circumferential) winding** - angle ≈ 90° to the axis; carries hoop stress.
- **Helical winding** - angle 15°-75°; carries both hoop and axial stress.
- **Polar winding** - fibre passes tangent to the polar openings; for spherical / dome vessels.

8.2 Steps, advantages and limitations

- Steps: fibre creel → resin bath (wet winding) or prepreg tow → delivery eye on carriage → winding on mandrel → cure (oven / rotating) → **mandrel extraction** → finishing.
- **Advantages**: high V_f (60-70 %), precise fibre orientation, automation, low cost of fibre (roving), high strength-to-weight.
- **Limitations**: only convex / axisymmetric shapes, mandrel cost, poor external surface, fibre cannot be laid along the axis easily.

Netting analysis (helical angle)

> For a closed pressure vessel: hoop stress $\sigma_\theta = pr / t$, axial $\sigma_x = pr / 2t$.

> Fibre-only theory gives the helical angle $\alpha = \tan^{-1}\sqrt{2} \approx$ **54.7°** from the axis for equal load sharing.

IX. Pultrusion and Pulforming

Pultrusion is a continuous process to make constant-cross-section profiles (rods, channels, I-beams, tubes). Fibres are **pulled** (not pushed) through a resin bath and then through a **heated die** where the resin cures; a puller grips the cured profile and a saw cuts to length.

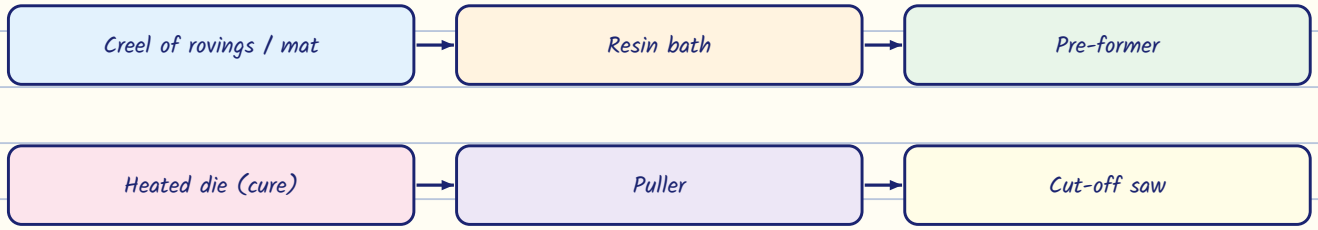


Fig. 7 Pultrusion line

- **Advantages**: continuous, fully automatic, low labour, V_f up to 70 %, excellent axial strength, low scrap.
- **Limitations**: only constant cross-section; heated die cost; low transverse strength unless mat is used.
- Uses: ladders, window frames, cable trays, bridge decks, fishing rods, electrical insulators, antenna poles.

9.1 Pulforming

A variation of pultrusion for **non-uniform / curved profiles**: after impregnation, the fibre profile is pulled through a **moving, closed mould** (or a die and a forming station) that shapes it into a curved or varying section. Used for leaf springs, curved handles, automotive parts.

X. Thermoforming, Injection and Blow Moulding

These are the main processes for **thermoplastic** matrix composites and filled plastics.

10.1 Thermoforming

- A heated **thermoplastic composite sheet** (organosheet) is softened, then shaped over / into a mould by **vacuum, pressure or matched dies** and cooled to set.
- Fast (30-60 s), recyclable; used for automotive panels, trays, aircraft interior, helmets.

10.2 Injection moulding

- Pellets of **short-fibre-filled thermoplastic** are melted in a heated barrel and **injected under high pressure** by a reciprocating screw into a closed, cooled mould; part is ejected after cooling.
- Cycle time seconds; very good for mass production of complex small parts (gears, housings, connectors).
- Limitations: fibre breakage → short fibre ($L < 1$ mm), low strength, anisotropy due to flow, expensive moulds.

10.3 Blow moulding

- A hot hollow tube of thermoplastic (**parison**) is clamped in a mould and inflated by compressed air against the mould walls, then cooled.
- Used for bottles, containers, fuel tanks and ducts; reinforced grades use short glass fibres or nano-fillers.

Process	Matrix	Product	Production
Hand lay-up	Thermoset	Boat, tank	Low
Bag moulding / autoclave	Thermoset prepreg	Aircraft skin	Low-medium
Filament winding	Thermoset	Pipes, vessels	Medium
Pultrusion	Thermoset	Profiles, rods	High (continuous)
Thermoforming	Thermoplastic	Panels, trays	High
Injection / blow	Thermoplastic	Small parts, bottles	Very high

XI. Quick Revision - Unit I

Important points

> Composite = matrix + reinforcement; **classification**: fibrous, laminated, particulate.
> Advantages: high specific strength / stiffness, tailoring, fatigue and corrosion resistance. Limitations: cost, brittleness, repair.
> Open mould = hand lay-up, spray-up; closed mould = compression, injection, RTM, pultrusion.
> Bag moulding: vacuum / pressure / autoclave; stack = release, peel ply, bleeder, breather, bag.
> Filament winding: hoop / helical / polar → pressure vessels; pultrusion: continuous profiles; pulforming: curved profiles.

Likely exam questions

> Define composite materials. Classify and compare fibrous, laminated and particulate composites.
> Explain hand lay-up technique with a neat sketch, merits and demerits.
> Describe structural laminate bag moulding and its production procedure.
> With neat sketches explain filament winding and pultrusion. Write short note on pulforming.
> Write applications of composites in automobile, aircraft, marine and sports.

Short Answer Questions (2 marks)

Q1. Define a composite material.

Ans: A material made of two or more constituents differing in form / composition, combined on a macroscopic scale, with properties better than the individual ones.

Q2. Name the two phases of a composite.

Ans: Continuous **matrix** and dispersed **reinforcement**.

Q3. What is specific strength?

Ans: Strength divided by density (σ/ρ); specific modulus = E/ρ.

Q4. Differentiate thermoset and thermoplastic matrix.

Ans: Thermoset cross-links irreversibly on curing (epoxy, polyester); thermoplastic softens on heating and can be reshaped (PEEK, nylon).

Q5. What is a prepreg?

Ans: Fibres (UD or fabric) pre-impregnated with partially cured resin; stored cold and cured in an autoclave.

Q6. Purpose of gel coat?

Ans: Smooth, pigmented, weather and water resistant outer surface in hand lay-up.

Q7. Why is a bleeder cloth used in bag moulding?

Ans: Absorbs excess resin so that fibre volume fraction increases and voids are reduced.

Q8. Which process makes pipes and pressure vessels?

Ans: Filament winding.

Q9. What is pultrusion?

Ans: Continuous pulling of resin-wet fibres through a heated die to form constant cross-section profiles.

Q10. Difference between pultrusion and pulforming?

Ans: Pulforming makes curved / varying cross-sections using a moving mould; pultrusion makes straight constant sections.

Q11. Give two limitations of composites.

Ans: High cost, brittleness / poor impact resistance, difficult repair and recycling, moisture absorption.

Glossary of Key Terms

Term	Meaning
Prepreg	Pre-impregnated fibre sheet with B-stage resin
Roving	Bundle of continuous untwisted strands wound on a spool
Mat	Randomly oriented chopped or continuous strand sheet
Debulking	Compacting plies under vacuum to remove air before cure
Gel time	Time for liquid resin to turn to a gel after catalyst is added
Autoclave	Heated pressure vessel for curing aerospace laminates
SMC / BMC	Sheet / bulk moulding compound for compression moulding
RTM	Resin transfer moulding: resin injected into a closed mould containing dry fibres

Answer Outlines for 10-mark Questions

Hand lay-up technique

- Sketch: mould, gel coat, plies, roller.
- Steps: release agent → gel coat → mat + resin → rolling → cure → demould.
- Advantages: low tooling cost, large parts; disadvantages: low V_f, operator dependent.
- Applications: boats, tanks, bath-tubs.

Filament winding

- Sketch of mandrel, carriage, resin bath, creel.
- Patterns: hoop, helical, polar; winding angle selection (54.7°).
- Cure, mandrel extraction; advantages and limitations; uses.

Bag moulding

- Lay-up stack and sketch of vacuum bag.
- Vacuum, pressure and autoclave variants with pressures.
- Production procedure steps; merits: higher V_f, lower voids.

UNIT II

Micro & Macro Mechanics of a Lamina

Rule of mixtures | four elastic moduli | Hooke's law | compliance & stiffness | angle lamina

I. Micro Mechanical Analysis of a Lamina

Micromechanics studies the behaviour of a lamina by considering the **individual properties and interaction of fibre and matrix**. Its aim is to predict the lamina (ply) properties E_1, E_2, ν_{12}, G_{12} from fibre and matrix properties and the volume fractions.

Term	Meaning
Lamina (ply)	Single flat layer of unidirectional / woven fibres in a matrix; basic building block, thickness 0.1-0.25 mm
Laminate	Stack of laminae bonded together at different orientations (Unit III)
Unidirectional (UD)	All fibres parallel; highest E_1, strongly orthotropic
Woven / fabric	Fibres in warp & weft; nearly equal strength in 2 directions (balanced weave)
Principal material axes	1 = along fibres, 2 = in-plane transverse, 3 = thickness

Approaches to find lamina properties

> **Mechanics of materials** (rule of mixtures) - simple assumptions on strain / stress; used in this unit.

> **Elasticity** (bounds, cylindrical assembly, Hashin) and **semi-empirical** (Halpin-Tsai) models.

> **Experiments** (tension, compression, V-notch shear tests) remain final authority.

I.1 Basic assumptions

- Lamina is macroscopically homogeneous, linearly elastic and **orthotropic**.
- Fibres are uniform, parallel and continuous; perfect fibre-matrix bond (no slip).
- Matrix is void-free, isotropic and linearly elastic; fibres also linearly elastic.
- Initially stress-free lamina; loads applied in principal material directions.

I.2 Volume fraction, weight fraction, density

Let v = volume, w = weight, ρ = density; subscripts c, f, m = composite, fibre, matrix.

$$V_f = v_f / v_c \,,\; V_m = v_m / v_c \,,\; V_f + V_m + V_v = 1$$
$$w_f = w_f / w_c \,,\; w_f + w_m = 1$$
$$\rho_c = \rho_f V_f + \rho_m V_m \,,\; 1/\rho_c = w_f/\rho_f + w_m/\rho_m$$
$$w_f = \rho_f V_f / \rho_c \,,\; V_f = (w_f/\rho_f) / (w_f/\rho_f + w_m/\rho_m)$$

Voids: $V_v = (\rho_t - \rho_e) / \rho_t$, where ρ_t = theoretical and ρ_e = experimental density. Even 1 % voids can cut inter-laminar shear strength by about 7 %.

II. Evaluation of the Four Elastic Moduli (Rule of Mixtures)

A lamina has four independent in-plane constants: E_1 (longitudinal modulus), E_2 (transverse modulus), ν_{12} (major Poisson's ratio) and G_{12} (in-plane shear modulus). They are found by the **strength-of-materials approach**.

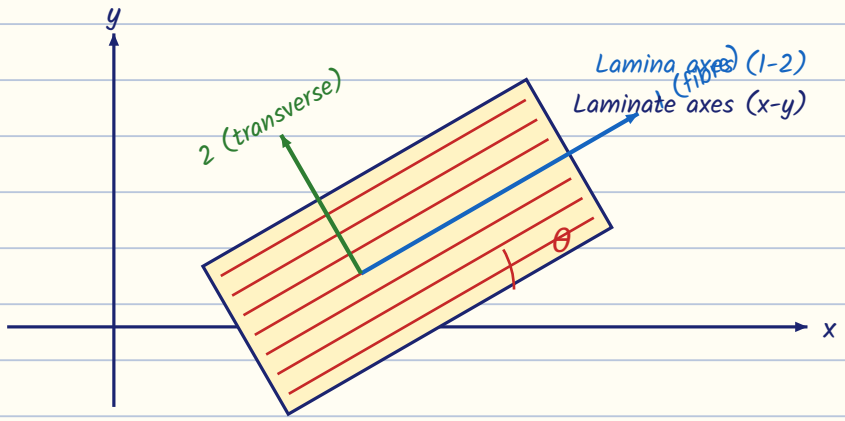
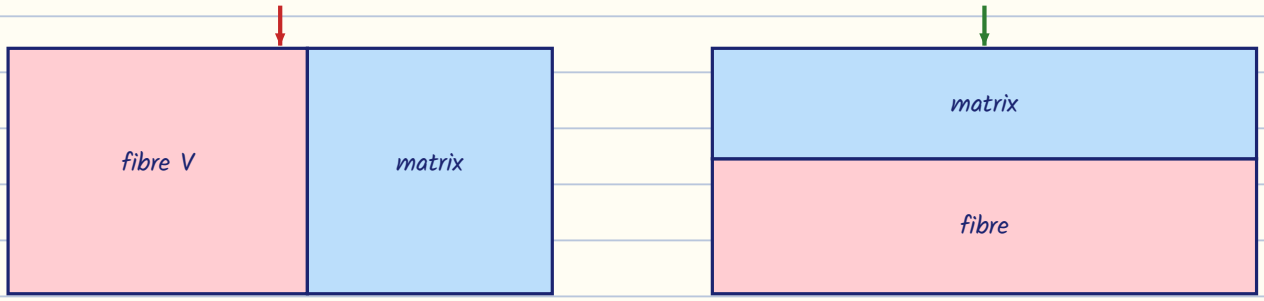


Fig. 1 Lamina with principal material axes 1-2



Voigt: equal strain → E1 **Reuss: equal stress → E2**

Fig. 2 Mechanics models: parallel (E1, 12) and series (E2, G12)

2.1 Longitudinal modulus E1 (equal strain model)

Load P is applied along the fibres. Fibre and matrix **strain equally**: $\varepsilon_c = \varepsilon_f = \varepsilon_m$. The load is shared: $P_c = P_f + P_m$.

$$\sigma_c A_c = \sigma_f A_f + \sigma_m A_m \Rightarrow E_c \varepsilon A_c = E_f \varepsilon A_f + E_m \varepsilon A_m$$
$$E_1 = E_f V_f + E_m V_m$$

E_1 is the **upper bound** (Voigt) and is very accurate (within 1-2 % of tests). Fibres carry the fraction $P_f/P_c = E_f V_f / E_1$.

Derivation of E1 - step by step

- Take a lamina element of length l, area $A_c = A_f + A_m$; volume fractions $V_f = A_f/A_c$, $V_m = A_m/A_c$ (length equal).
- Equilibrium: $P_c = P_f + P_m \rightarrow \sigma_1 A_c = \sigma_f A_f + \sigma_m A_m \rightarrow \sigma_1 = \sigma_f V_f + \sigma_m V_m$.
- Compatibility: the fibre, matrix and composite elongate equally, $\varepsilon_1 = \varepsilon_f = \varepsilon_m$.
- Constitutive: $\sigma_f = E_f \varepsilon_1$, $\sigma_m = E_m \varepsilon_1$; substitute $\rightarrow \sigma_1 = (E_f V_f + E_m V_m)\, \varepsilon_1 \rightarrow E_1 = \sigma_1/\varepsilon_1$.

2.2 Transverse modulus E2 (equal stress model)

Load is applied **perpendicular** to fibres. Both constituents carry the **same stress** σ, but strain differently; the total deformation is the sum: $\delta_c = \delta_f + \delta_m$, i.e. $\varepsilon_2 t_c = \varepsilon_f t_f + \varepsilon_m t_m$.

$$\sigma/E_2 = V_f\, \sigma/E_f + V_m\, \sigma/E_m$$
$$1/E_2 = V_f/E_f + V_m/E_m \text{ or } E_2 = E_f E_m / (E_f V_m + E_m V_f)$$

This is the **lower bound** (Reuss). It under-predicts test values; the **Halpin-Tsai** semi-empirical equation ($E_2/E_m = (1 + \xi\eta V_f)/(1 - \eta V_f)$, $\eta = (E_f/E_m - 1)/(E_f/E_m + \xi)$) gives a better fit.

2.3 Major Poisson's ratio 12

Poisson's ratio = - (transverse strain)/(longitudinal strain) when loaded along fibres. Transverse deformation of the lamina is the sum of those of fibre and matrix (equal longitudinal strain ε_1):

$$\nu_{12} = \nu_f V_f + \nu_m V_m \text{ and } \nu_{21} = \nu_{12}\, E_2/E_1$$

2.4 In-plane shear modulus G12

Shear stress τ acts on fibre and matrix equally (series model) and shear deformations add: $\gamma_c t_c = \gamma_f t_f + \gamma_m t_m$.

$$1/G_{12} = V_f/G_f + V_m/G_m \Rightarrow G_{12} = G_f G_m / (G_f V_m + G_m V_f)$$
with $G = E / 2(1 + \nu)$ for each isotropic constituent

Constant	Rule of mixtures	Remark
E_1	$E_f V_f + E_m V_m$	Voigt; fibre dominated; accurate
E_2	$E_f E_m / (E_f V_m + E_m V_f)$	Reuss; matrix dominated; low

Constant	Rule of mixtures	Remark
ν_{12}	$\nu_f V_f + \nu_m V_m$	$\nu_{21} = \nu_{12} E_2 / E_1$
G_{12}	$G_f G_m / (G_f V_m + G_m V_f)$	Matrix dominated; under-predicts

2.5 Effect of fibre volume fraction

V_f	0.2	0.3	0.4	0.5	0.6	0.7
E_1 (GPa)	49.4	72.1	94.8	116.8	139.4	162.1
E_2 (GPa)	4.3	4.8	5.7	6.9	8.6	11.4

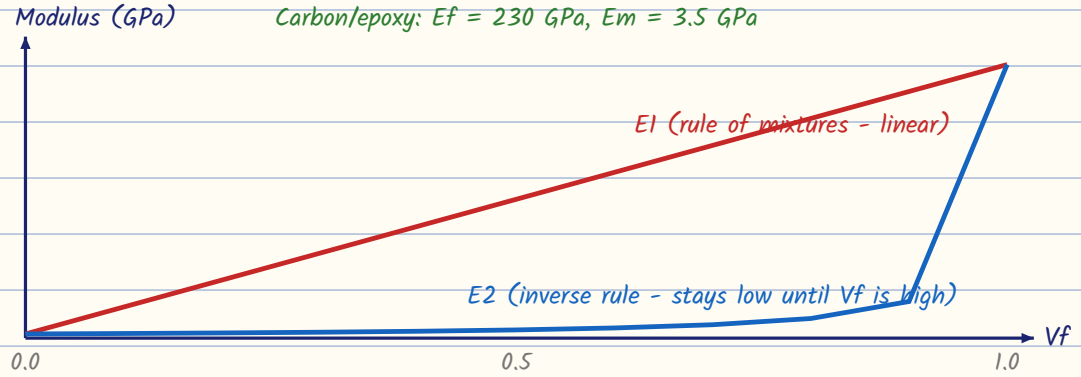


Fig. 3 Variation of E1 and E2 with Vf (carbon/epoxy)

- E_1 varies **linearly** with V_f; to double E_1 we need to double V_f.
- E_2 is controlled by the **matrix**: it rises only slowly until V_f is large - so transverse stiffness cannot be bought with more fibre.
- Practical limit $V_f \approx 0.65$–0.70 (hexagonal packing limit 0.907, square 0.785).
- Halpin-Tsai with $\xi = 2$ for E_2 gives a more realistic (higher) value than the inverse rule.

III. Numerical Problems on Micromechanics

Problem 1

Glass/epoxy lamina: $E_f = 72$ GPa, $E_m = 3.5$ GPa, $G_f = 30$ GPa, $G_m = 1.3$ GPa, $\nu_f = 0.22$, $\nu_m = 0.35$, $V_f = 0.6$. Find E_1, E_2, ν_{12}, G_{12}.

$V_m = 1 - 0.6 = 0.4$

$E_1 = 72(0.6) + 3.5(0.4) = 43.2 + 1.4 = \mathbf{44.6\ GPa}$

$E_2 = (72 \times 3.5) / (72 \times 0.4 + 3.5 \times 0.6) = 252 / 30.9 = \mathbf{8.16\ GPa}$

$\nu_{12} = 0.22(0.6) + 0.35(0.4) = 0.132 + 0.14 = \mathbf{0.272}$

$G_{12} = (30 \times 1.3) / (30 \times 0.4 + 1.3 \times 0.6) = 39 / 12.78 = \mathbf{3.05\ GPa}$

Problem 2

A glass/epoxy composite has weight fraction of glass $W_f = 0.7$, $\rho_f = 2.5$ g/cc, $\rho_m = 1.2$ g/cc. Find V_f and ρ_c.

$1/\rho_c = 0.7/2.5 + 0.3/1.2 = 0.28 + 0.25 = 0.53 \Rightarrow \rho_c =$ **1.887 g/cc**

$V_f = \rho_c W_f/\rho_f = 1.887 \times 0.7 / 2.5 =$ **0.528** (52.8 %)

Problem 3

Carbon/epoxy: $E_f = 230$ GPa, $E_m = 3.5$ GPa, $V_f = 0.5$. Find E_1, E_2 and the load share of fibres along 1.

$E_1 = 230(0.5) + 3.5(0.5) =$ **116.75 GPa**; $E_2 = 230 \times 3.5 / (230 \times 0.5 + 3.5 \times 0.5) = 805 / 116.75$ = **6.90 GPa**

Fibre load share = $E_f V_f/E_1 = 115/116.75 =$ **98.5 %** ($E_1/E_2 \approx 17$: strongly orthotropic).

Problem 7

Carbon/epoxy ($E_f = 230$, $E_m = 3.5$ GPa). What V_f is needed for $E_1 = 100$ GPa? What strain does $\sigma_1 = 100$ MPa produce?

$100 = 230\ V_f + 3.5(1 - V_f) \Rightarrow V_f = 96.5 / 226.5 =$ **0.426**

$\varepsilon_1 = \sigma_1/E_1 = 100$ MPa / 100 GPa = **1000** $\mu\varepsilon$ (0.1 %) ; fibre stress = $E_f\varepsilon_1 =$ **230 MPa**, matrix stress = **3.5 MPa**.

Problem 8

A lamina has $E_1 = 140$ GPa, $E_2 = 10$ GPa, $\nu_{12} = 0.3$. Find ν_{21} and check the stability condition.

$\nu_{21} = \nu_{12}E_2/E_1 = 0.3 \times 10/140 =$ **0.0214**. Limit: $|\nu_{12}| < \sqrt{(E_1/E_2)} = \sqrt{14} = 3.74 \rightarrow 0.3 < 3.74$ ✓ (valid).

IV. Macro Mechanics of a Lamina

Macromechanics treats the lamina as a **homogeneous, anisotropic** continuum and uses average (apparent) properties E_1, E_2, ν_{12}, G_{12} to find stress-strain relations; the fibre and matrix are not separately visible.

4.1 Generalised Hooke's law

For a 3-D linearly elastic body the stress components σ_i ($\sigma_1 \ldots \sigma_3$ normal, τ_{23}, τ_{31}, τ_{12} shear) and strains ε_j are related by (contracted notation):

$$\{\sigma_i\} = [C_{ij}]\{\varepsilon_j\} \text{ (stiffness) } \{\varepsilon_i\} = [S_{ij}]\{\sigma_j\} \text{ (compliance)}$$
$$[S] = [C]^{-1} \; ; \; i, j = 1 \dots 6$$

The 81 constants of the 4th-order tensor reduce to **36** by symmetry of stress and strain, and to **21** (symmetric matrix $C_{ij} = C_{ji}$) by the existence of a **strain-energy function**.

Engineering constants are defined from simple tests: $E_i = \sigma_i/\varepsilon_i$ (tension), $\nu_{ij} = -\varepsilon_j/\varepsilon_i$ (load in i), $G_{ij} = \tau_{ij}/\gamma_{ij}$ (shear). The compliance matrix of an **orthotropic** material (9 constants, 12 non-zero terms, 3 pairs equal) has the form:

$$\begin{bmatrix} 1/E1 & -\nu21/E2 & -\nu31/E3 & 0 & 0 & 0 \\ -\nu12/E1 & 1/E2 & -\nu32/E3 & 0 & 0 & 0 \\ -\nu13/E1 & -\nu23/E2 & 1/E3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/G23 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/G31 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/G12 \end{bmatrix}$$

Orthotropic compliance [S]: normal-shear coupling terms are zero

4.2 Number of elastic constants for different materials

Material	Symmetry planes	Independent constants
Fully anisotropic (triclinic)	None	21
Monoclinic	One plane of symmetry	13
Orthotropic (lamina, wood)	Three mutually ⊥ planes	9 ($E_1, E_2, E_3, \nu_{12}, \nu_{13}, \nu_{23}, G_{12}, G_{13}, G_{23}$)
Transversely isotropic (UD fibres)	Isotropic in plane 2-3	5 ($E_1, E_2, \nu_{12}, \nu_{23}, G_{12}$)
Isotropic (metals)	Infinite planes	2 (E, ν); $G = E/2(1+\nu)$

Remember: In the orthotropic case **no coupling** exists between normal stresses and shear strains ($S_{16} = S_{26} = 0$) when loaded along the material axes.

4.3 Plane stress: two-dimensional relations

A thin lamina has $\sigma_3 = \tau_{23} = \tau_{31} = 0$ (**plane stress**), so only σ_1, σ_2, τ_{12} remain.

$$\{\varepsilon_1 \; \varepsilon_2 \; \gamma_{12}\}^T = [S]\{\sigma_1 \; \sigma_2 \; \tau_{12}\}^T$$
$$S_{11} = 1/E_1 \, , \; S_{22} = 1/E_2 \, , \; S_{12} = -\nu_{12}/E_1 = -\nu_{21}/E_2 \, , \; S_{66} = 1/G_{12}$$

Inverting gives the **reduced stiffness matrix** $[Q] = [S]^{-1}$:

$$Q_{11} = E_1/(1 - \nu_{12}\nu_{21}) \, , \; Q_{22} = E_2/(1 - \nu_{12}\nu_{21})$$
$$Q_{12} = \nu_{12}E_2/(1 - \nu_{12}\nu_{21}) \, , \; Q_{66} = G_{12}$$

Maxwell-Betti reciprocity: $\nu_{12}/E_1 = \nu_{21}/E_2$. Positive definiteness requires $E, G > 0$ and $|\nu_{12}| < \sqrt{(E_1/E_2)}$.

V. Hooke's Law for a Two-Dimensional Angle Lamina

When fibres make an angle θ with the loading axis x (**off-axis / angle lamina**), stresses and strains in x-y are related by the **transformed** matrices. Shear-extension coupling appears: a normal stress produces shear strain.

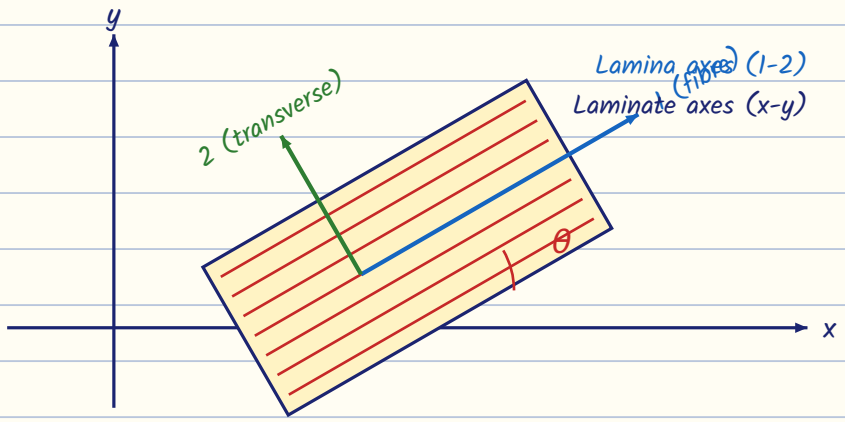


Fig. 4 Angle lamina: material axes (1,2) rotated by from x-y

5.1 Transformation of stresses and strains

$$\{\sigma_1\ \sigma_2\ \tau_{12}\}^T = [T]\ \{\sigma_x\ \sigma_y\ \tau_{xy}\}^T$$

$$[T] = [\ m^2\ n^2\ 2mn\ ;\ n^2\ m^2\ -2mn\ ;\ -mn\ mn\ m^2 - n^2\ ]$$

$$m = \cos\theta\ ,\ n = \sin\theta$$

Strains transform the same way with the engineering shear strain γ: use the Reuter matrix $[R] = \text{diag}(1, 1, 2)$, giving ε-transformation $[R][T][R]^{-1} = [T]^{-T}$.

Derivation idea (wedge element)

- Cut a small triangular element from the lamina with its inclined face normal to axis 1 (angle θ from x).
- Write force equilibrium along 1 and 2 using areas A·cosθ, A·sinθ → gives the [T] matrix ($\sigma_1 = \sigma_x m^2 + \sigma_y n^2 + 2\tau_{xy} mn$ ...).
- Strains transform by the same geometry but with tensor shear γ/2, hence the Reuter matrix [R].
- Substituting $\{\sigma\} = [Q]\{\varepsilon\}$: $\{\sigma_{xy}\} = [T]^{-1}[Q][T]^{-T}\{\varepsilon_{xy}\} = [\bar{Q}]\{\varepsilon_{xy}\}$.

5.2 Transformed reduced stiffness [Q] and compliance [S]

$$\{\sigma_x\ \sigma_y\ \tau_{xy}\}^T = [\bar{Q}]\ \{\varepsilon_x\ \varepsilon_y\ \gamma_{xy}\}^T,\quad [\bar{Q}] = [T]^{-1}[Q][R][T][R]^{-1}$$
$$\{\varepsilon_x\ \varepsilon_y\ \gamma_{xy}\}^T = [\bar{S}]\ \{\sigma_x\ \sigma_y\ \tau_{xy}\}^T,\quad [\bar{S}] = [\bar{Q}]^{-1}$$

Expanded forms ($m = \cos\theta$, $n = \sin\theta$):

$$\bar{Q}_{11} = Q_{11}m^4 + 2(Q_{12} + 2Q_{66})m^2n^2 + Q_{22}n^4$$
$$\bar{Q}_{12} = (Q_{11} + Q_{22} - 4Q_{66})m^2n^2 + Q_{12}(m^4 + n^4)$$
$$\bar{Q}_{22} = Q_{11}n^4 + 2(Q_{12} + 2Q_{66})m^2n^2 + Q_{22}m^4$$
$$\bar{Q}_{16} = (Q_{11} - Q_{12} - 2Q_{66})m^3n + (Q_{12} - Q_{22} + 2Q_{66})mn^3$$
$$\bar{Q}_{26} = (Q_{11} - Q_{12} - 2Q_{66})mn^3 + (Q_{12} - Q_{22} + 2Q_{66})m^3n$$
$$\bar{Q}_{66} = (Q_{11} + Q_{22} - 2Q_{12} - 2Q_{66})m^2n^2 + Q_{66}(m^4 + n^4)$$

Properties of [Q]

- Fully populated (9 non-zero terms) but only **4 independent** constants (+ θ).
- Symmetric: $\bar{Q}_{ij} = \bar{Q}_{ji}$.
- At $\theta = 0°$ or $90°$: $\bar{Q}_{16} = \bar{Q}_{26} = 0$ (specially orthotropic).
- $\bar{Q}_{16}$, $\bar{Q}_{26}$ are odd functions of θ: sign changes for $+\theta$ / $-\theta$.

5.3 Engineering constants of an angle lamina

$$1/E_x = m^4/E_1 + (1/G_{12} - 2\nu_{12}/E_1)\,m^2n^2 + n^4/E_2$$
$$1/E_y = n^4/E_1 + (1/G_{12} - 2\nu_{12}/E_1)\,m^2n^2 + m^4/E_2$$
$$1/G_{xy} = 2(2/E_1 + 2/E_2 + 4\nu_{12}/E_1 - 1/G_{12})\,m^2n^2 + (m^4 + n^4)/G_{12}$$
$$\nu_{xy} = E_x[\nu_{12}(m^4 + n^4)/E_1 - (1/E_1 + 1/E_2 - 1/G_{12})\,m^2n^2]$$

Coefficients of mutual influence: $\eta_{x,xy} = \varepsilon_{xy} / \varepsilon_x$ (shear strain per unit normal strain) and $\eta_{xy,x}$ - the sign of coupling follows θ.

VI. Numerical Problems - Angle Lamina

Problem 4

Glass/epoxy: $E_1 = 38.6$, $E_2 = 8.27$, $G_{12} = 4.14$ GPa, $\nu_{12} = 0.26$. Find [S], [Q] and [$\bar{Q}$] at $\theta = 45°$.

$\nu_{21} = 0.26 \times 8.27/38.6 = 0.0557$; $1 - \nu_{12}\nu_{21} = 0.9855$

[S] (1/GPa): $S_{11} = 0.02591$, $S_{12} = -0.00674$, $S_{22} = 0.12092$, $S_{66} = 0.24155$

[Q] (GPa): $Q_{11} = 39.17$, $Q_{12} = 2.18$, $Q_{22} = 8.39$, $Q_{66} = 4.14$

At 45° ($m^2 = n^2 = 0.5$): $\bar{Q}_{11} = \bar{Q}_{22} = 17.12$, $\bar{Q}_{12} = 8.84$, $\bar{Q}_{66} = 10.80$, $\bar{Q}_{16} = \bar{Q}_{26} = 7.69$ GPa.

θ	$\bar{Q}_{11}$	$\bar{Q}_{12}$	$\bar{Q}_{22}$	$\bar{Q}_{16}$	$\bar{Q}_{26}$	$\bar{Q}_{66}$
30°	26.48	7.18	11.09	9.55	3.78	9.13
45°	17.12	8.84	17.12	7.69	7.69	10.80
60°	11.09	7.18	26.48	3.78	9.55	9.13

Problem 5

Same lamina, stresses in x-y: $\sigma_x = 2$ MPa, $\sigma_y = -3$ MPa, $\tau_{xy} = 4$ MPa, $\theta = 30°$. Find stresses and strains in material axes.

$m = \cos 30° = 0.866$, $n = 0.5$. $\sigma_1 = m^2\sigma_x + n^2\sigma_y + 2mn\,\tau_{xy} = 1.5 - 0.75 + 3.464 =$ **4.214 MPa**

$\sigma_2 = n^2\sigma_x + m^2\sigma_y - 2mn\,\tau_{xy} = 0.5 - 2.25 - 3.464 =$ **-5.214 MPa**

$\tau_{12} = -mn\,\sigma_x + mn\,\sigma_y + (m^2 - n^2)\,\tau_{xy} = -0.866 - 1.299 + 2.0 =$ **-0.165 MPa**

$\varepsilon_1 = S_{11}\sigma_1 + S_{12}\sigma_2 =$ **144.3** $\mu\varepsilon$; $\varepsilon_2 = S_{12}\sigma_1 + S_{22}\sigma_2 =$ **-658.9** $\mu\varepsilon$; $\gamma_{12} = S_{66}\tau_{12} =$ **-39.9** μ

Problem 6

For the same material find E_x, E_y, G_{xy} at $\theta = 30°$ and 45°.

$\theta = 30°$: $E_x =$ **15.41 GPa**, $E_y =$ **8.90 GPa**, $G_{xy} =$ **5.54 GPa**

$\theta = 45°$: $E_x = E_y =$ **10.67 GPa**, $G_{xy} =$ **6.24 GPa** (E_x falls from 38.6 to 10.7 GPa - fibre direction matters!)

Problem 9

Glass/epoxy lamina at $\theta = 60°$ is subjected to strains $\varepsilon_x = 0.001$, $\varepsilon_y = 0$, $\gamma_{xy} = 0$. Find the stresses σ_x, σ_y, τ_{xy}.

At 60°: $\bar{Q}_{11} = 11.09$, $\bar{Q}_{12} = 7.18$, $\bar{Q}_{16} = 3.78$ GPa (from the table above).

$\sigma_x = \bar{Q}_{11}\varepsilon_x =$ **11.09 MPa** ; $\sigma_y = \bar{Q}_{12}\varepsilon_x =$ **7.18 MPa** ; $\tau_{xy} = \bar{Q}_{16}\varepsilon_x =$ **3.78 MPa**

Observe: A pure normal strain ε_x induces **shear stress** τ_{xy} - a result of anisotropy; it does not occur in isotropic materials.

VII. Stress-Strain Relations for a Lamina of Arbitrary Orientation

- **Step 1** - find E_1, E_2, ν_{12}, G_{12}; compute [S] and [Q].
- **Step 2** - set $m = \cos\theta$, $n = \sin\theta$ (θ +ve counter-clockwise from x to fibre direction 1).
- **Step 3** - get $[\bar{Q}]$ by formula, or $[S] = [\bar{Q}]^{-1}$.
- **Step 4** - given σ_x, σ_y, $\tau_{xy} \rightarrow \varepsilon_x$, ε_y, $\gamma_{xy} = [S]\{\sigma\}$; or given strains → stresses.
- **Step 5** - transform to material axes ($\{\sigma_{1,2,12}\} = [T]\{\sigma_{xy}\}$) for failure checking (Unit III).

Special behaviour of angle lamina

- **Shear-extension coupling**: σ_x alone produces γ_{xy} ($S_{16} \neq 0$); a rectangular element distorts.
- **Normal-shear (η) interaction** is absent for $\theta = 0°$, $90°$, and for isotropic materials.
- E_x varies from E_1 ($\theta = 0°$) to E_2 ($\theta = 90°$): minimum may be below E_2 when G_{12} is small.

Specially orthotropic ($\theta = 0°, 90°$)	Generally orthotropic (angle lamina)
Axes x-y coincide with 1-2	Axes x-y rotated by θ from 1-2
$\bar{Q}_{16} = \bar{Q}_{26} = 0$: no shear coupling	$\bar{Q}_{16}, \bar{Q}_{26} \neq 0$: shear-extension coupling
4 constants: E_1, E_2, ν_{12}, G_{12}	4 constants + θ, with 6 non-zero $\bar{Q}_{ij}$
Used in micromechanics and tests	Used in laminate analysis (Unit III)

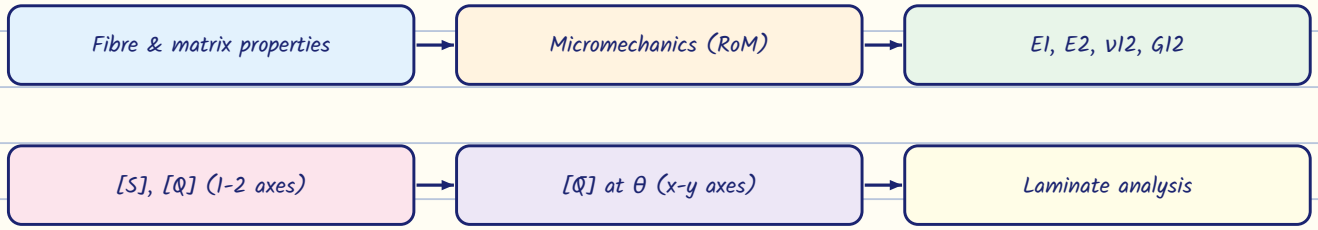


Fig. 5 Road-map from constituents to laminate

7.1 Typical unidirectional lamina properties (V_f 0.5–0.6)

Material	E_1 (GPa)	E_2 (GPa)	G_{12} (GPa)	ν_{12}
E-glass / epoxy	38.6	8.27	4.14	0.26
Kevlar-49 / epoxy	76	5.5	2.3	0.34
Carbon T300 / epoxy	181	10.3	7.17	0.28
Boron / epoxy	204	18.5	5.59	0.23

Problem 10

Carbon/epoxy ($E_1 = 181$, $E_2 = 10.3$, $G_{12} = 7.17$ GPa, $\nu_{12} = 0.28$): find E_x for fibre angles 15°, 30°, 45°, 60°.

Use $1/E_x = m^4/E_1 + (1/G_{12} - 2\nu_{12}/E_1)m^2n^2 + n^4/E_2$

θ	0°	15°	30°	45°	60°	90°
E_x (GPa)	181	72.6	28.8	16.7	12.4	10.3

Conclusion: a mere **15° misalignment** cuts the stiffness by 60 % - fibre alignment during manufacture is critical (and the reason for multi-angle laminates).

Problem II (Halpin-Tsai)

Find E_2 for carbon/epoxy (E_f = 230, E_m = 3.5 GPa, V_f = 0.5) with ξ = 2 and compare with the inverse rule of mixtures.

$\eta = (E_f/E_m - 1) / (E_f/E_m + \xi) = (65.71 - 1) / (65.71 + 2) =$ **0.956**

$E_2 = E_m(1 + \xi\eta V_f) / (1 - \eta V_f) = 3.5(1 + 0.956) / (1 - 0.478) =$ **13.1 GPa** vs 6.9 GPa (inverse RoM): Halpin-Tsai is nearly twice - closer to tests.

Viva / short questions

> Why is E_1 called the upper bound and E_2 the lower bound?
> What is the difference between a lamina and a laminate?
> How many independent constants for a UD lamina in plane stress? (4: E_1, E_2, ν_{12}, G_{12})
> Why does [Q] have 9 non-zero terms but only 4 independent constants?
> Why is $\nu_{21} \neq \nu_{12}$ for an orthotropic lamina?

VIII. Quick Revision - Unit II

Formulae to remember

> $E_1 = E_fV_f + E_mV_m$; $1/E_2 = V_f/E_f + V_m/E_m$; $\nu_{12} = \nu_fV_f + \nu_mV_m$; $1/G_{12} = V_f/G_f + V_m/G_m$
> Orthotropic: 9 constants; transversely isotropic: 5; isotropic: 2; anisotropic: 21.
> $[S] = [Q]^{-1}$; $[\bar{Q}] = [T]^{-1}[Q][R][T][R]^{-1}$; engineering constants E_x, E_y, G_{xy}, ν_{xy} vs θ.

Likely exam questions

> Derive E_1, E_2, ν_{12} and G_{12} using the rule of mixtures.
> Explain Hooke's law for different types of materials; number of independent elastic constants.
> Derive 2-D relationship of compliance and stiffness matrices for a lamina.
> Derive stress-strain relations for a lamina of arbitrary orientation; numerical problems.

Short Answer Questions (2 marks)

Q1. State the rule of mixtures for E_1.

Ans: $E_1 = E_fV_f + E_mV_m$ (equal-strain, upper bound).

Q2. Write the expression for the transverse modulus.

Ans: $1/E_2 = V_f/E_f + V_m/E_m$ (equal-stress, lower bound).

Q3. How many independent elastic constants are needed for an orthotropic lamina in plane stress?

Ans: Four: E_1, E_2, ν_{12}, G_{12}.

Q4. Define ν_{12}.

Ans: Ratio of transverse strain (direction 2) to longitudinal strain (direction 1), with sign change, when loaded in direction 1.

Q5. What is the physical meaning of $\bar{Q}_{16} \neq 0$?

Ans: Shear-extension coupling: normal strain produces shear stress in an off-axis lamina.

Q6. What is the Reuter matrix?

Ans: $[R] = \text{diag}(1, 1, 2)$ converting tensor shear strain to engineering shear strain.

Q7. Why is $[Q] \neq [S]^{-1}$ in 3-D but equal in plane stress?

Ans: In plane stress, the reduced stiffness $[Q]$ is the inverse of the 3×3 in-plane compliance $[S]$; the 3-D $[C]$ is a 6×6 matrix.

Q8. Number of constants in isotropic and anisotropic materials?

Ans: 2 and 21 respectively.

Q9. At which angle is the shear modulus G_{xy} of a lamina maximum?

Ans: Around 45°.

Practice Problems with Answers

Problem A

Carbon/epoxy $V_f = 0.6$: $E_f = 230$, $E_m = 3.5$ GPa, $\nu_f = 0.2$, $\nu_m = 0.35$. Find E_1, E_2, ν_{12}.

$E_1 = 230(0.6) + 3.5(0.4) =$ **139.4 GPa** ; $E_2 = 230 \times 3.5/(230 \times 0.4 + 3.5 \times 0.6) = 805/94.1 =$ **8.55 GPa** ; $\nu_{12} = 0.2(0.6) + 0.35(0.4) =$ **0.26**.

Problem B

Epoxy ($\rho = 1.2$) with E-glass ($\rho = 2.5$): fibre weight fraction 0.5. Find V_f.

$V_f = (0.5/2.5)/(0.5/2.5 + 0.5/1.2) = 0.2/(0.2 + 0.4167) =$ **0.324**.

UNIT III

Biaxial Strength Theories & Laminate Analysis

Max stress | max strain | Tsai-Hill | Tsai-Wu | Kirchhoff hypothesis | CLT | A, B, D matrices

I. Biaxial Strength Theories of a Lamina

A lamina is usually under a **combined (biaxial) stress state** σ_1, σ_2, τ_{12} in its material axes. A **failure criterion** predicts whether the lamina fails, by comparing this state with the basic strengths obtained from simple tests. Because the lamina is **anisotropic**, strength depends on direction and on the sign of stress (tension ≠ compression).

I.1 The five basic strengths of a UD lamina

Symbol	Strength	Meaning
X_t (X)	Longitudinal tensile	along fibres, tension
X_c (X′)	Longitudinal compressive	along fibres, compression (microbuckling)
Y_t (Y)	Transverse tensile	⊥ fibres, tension (matrix / interface)
Y_c (Y′)	Transverse compressive	⊥ fibres, compression
S	In-plane shear	τ_{12} at failure

Material	X_t	X_c	Y_t	Y_c	S (MPa)
Glass/epoxy	1062	610	31	118	72
Carbon/epoxy (T300)	1500	1500	40	246	68
Kevlar/epoxy	1400	235	12	53	34

Note: $Y \ll X$: a lamina is **20–40 times weaker** across the fibres - the matrix cracks first (first-ply failure).

I.2 General procedure

- Find the applied stresses σ_x, σ_y, τ_{xy} (global axes).
- Transform to material axes: $\{\sigma_1, \sigma_2, \tau_{12}\}^T = [T]\{\sigma_x, \sigma_y, \tau_{xy}\}^T$.
- Apply the criterion using the five strengths; failure if the condition is met or exceeded.
- Strength ratio R = (failure stress)/(applied stress); **R > 1 safe**, R = 1 failure, R < 1 fails.

II. Maximum Stress Theory

Failure occurs when **any one** stress component in the material axes reaches its corresponding strength. The components act **independently** (no interaction).

$$-X_c < \sigma_1 < X_t \; ; \; -Y_c < \sigma_2 < Y_t \; ; \; |\tau_{12}| < S$$

If **any** inequality is violated the lamina fails, and the mode of failure is identified directly (fibre failure, matrix failure or shear).

2.1 Off-axis uniaxial tension x

For a lamina loaded by σ_x at angle θ ($\sigma_1 = \sigma_x \cos^2\theta$, $\sigma_2 = \sigma_x \sin^2\theta$, $\tau_{12} = -\sigma_x \sin\theta\cos\theta$), the strength is the **smallest** of:

$$\sigma_x = X_t \,/\, \cos^2\theta \text{ (fibre failure)}$$
$$\sigma_x = Y_t \,/\, \sin^2\theta \text{ (transverse failure)}$$
$$\sigma_x = S \,/\, (\sin\theta\cos\theta) \text{ (shear failure)}$$

- Predicts **abrupt kinks** in strength-vs-angle curve where the failure mode changes.
- Simple, mode-specific; but ignores **interaction** of stresses, so it is often unconservative near mixed states.

III. Maximum Strain Theory

Failure occurs when **any strain component** in the material axes reaches its ultimate value. It allows for **Poisson interaction** because strains depend on all stresses.

$$-\varepsilon_{1c} < \varepsilon_1 < \varepsilon_{1t} \; ; \; -\varepsilon_{2c} < \varepsilon_2 < \varepsilon_{2t} \; ; \; |\gamma_{12}| < \gamma_{12u}$$
$$\varepsilon_{1t} = X_t/E_1 \; , \; \varepsilon_{2t} = Y_t/E_2 \; , \; \gamma_{12u} = S/G_{12}$$

With plane stress: $\varepsilon_1 = (\sigma_1 - \nu_{12}\sigma_2)/E_1$, $\varepsilon_2 = (\sigma_2 - \nu_{21}\sigma_1)/E_2$, $\gamma_{12} = \tau_{12}/G_{12}$. For off-axis tension:

$$\sigma_x = X_t \,/\, (\cos^2\theta - \nu_{12} \sin^2\theta) \; , \; \sigma_x = Y_t \,/\, (\sin^2\theta - \nu_{21} \cos^2\theta) \; , \; \sigma_x = S/(\sin\theta \cos\theta)$$

Compare: Max stress and max strain coincide only when $\nu_{12} = 0$. Both are **non-interactive** criteria - each mode is checked separately.

IV. Tsai-Hill Theory (Interactive)

Hill (1948) extended the **von Mises** yield criterion to anisotropic materials; Tsai adapted it to composites. A **single quadratic equation** combines all stress components, so one stress lowers the allowable value of the others (**interaction**). For a plane-stress lamina:

$$\sigma_1^2 \,/\, X^2 - \sigma_1\sigma_2 \,/\, X^2 + \sigma_2^2 \,/\, Y^2 + \tau_{12}^2 \,/\, S^2 = 1$$

- Use $X = X_t$ if $\sigma_1 > 0$ else X_c; $Y = Y_t$ if $\sigma_2 > 0$ else Y_c (strengths chosen by sign of stress).
- Failure if LHS ≥ 1; **strength ratio** $R = 1/\sqrt{(LHS)}$.

- Does not distinguish failure mode; assumes equal tension and compression strengths in each direction.
- Off-axis tension: $1/\sigma_x^2 = \cos^4\theta/X^2 + (1/S^2 - 1/X^2)\sin^2\theta\cos^2\theta + \sin^4\theta/Y^2$ (smooth curve).

Origin of Tsai-Hill (Hill, 1948)

> Hill's anisotropic yield function: $F(\sigma_2 - \sigma_3)^2 + G(\sigma_3 - \sigma_1)^2 + H(\sigma_1 - \sigma_2)^2 + 2L\tau_{23}^2 + 2M\tau_{31}^2 + 2N\tau_{12}^2 = 1$.

> Apply to simple tests: $\sigma_1 = X$ alone → $G + H = 1/X^2$; $\sigma_2 = Y$ → $F + H = 1/Y^2$; $\sigma_3 = Z$ → $F + G = 1/Z^2$; $\tau_{12} = S$ → $2N = 1/S^2$.

> For a UD lamina (transversely isotropic, $Y = Z$, plane stress $\sigma_3 = 0$): $F + G$... gives $G = 1/(2X^2)\cdot(\ldots)$ and finally $H = 1/(2X^2)$, $G + H = 1/X^2$.

> Substituting → $\sigma_1^2/X^2 - \sigma_1\sigma_2/X^2 + \sigma_2^2/Y^2 + \tau_{12}^2/S^2 = 1$.

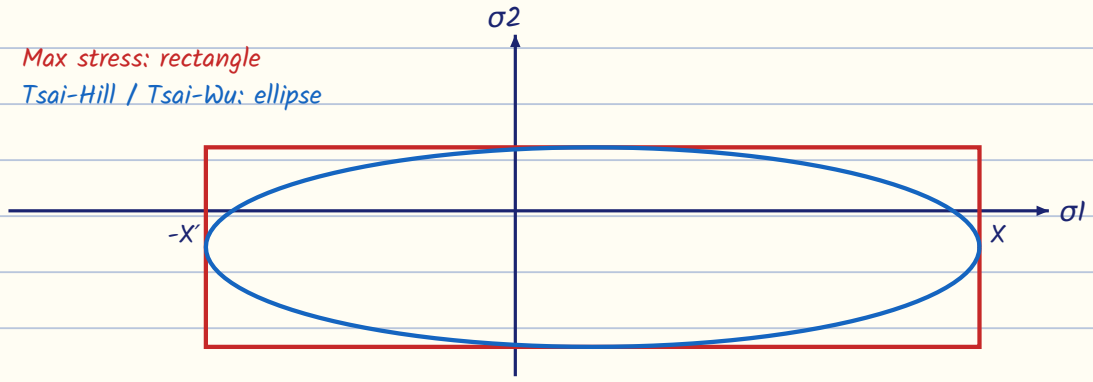


Fig. 1 Failure envelope in 1-2 space ($\tau_{12} = 0$): max stress vs Tsai-Hill / Tsai-Wu

V. Tsai-Wu Tensor Theory

Tsai and Wu (1971) proposed a **tensor polynomial** failure criterion that accounts for **different tensile and compressive strengths** and for interaction through a cross term F_{12}. It reduces to Tsai-Hill in a special case.

$$F_i\sigma_i + F_{ij}\sigma_i\sigma_j = 1 \quad (i, j = 1, 2, 6)$$

$$F_1\sigma_1 + F_2\sigma_2 + F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + F_{66}\tau_{12}^2 + 2F_{12}\sigma_1\sigma_2 = 1$$

$$F_1 = 1/X_t - 1/X_c \,,\; F_2 = 1/Y_t - 1/Y_c$$

$$F_{11} = 1/(X_tX_c) \,,\; F_{22} = 1/(Y_tY_c) \,,\; F_{66} = 1/S^2$$

$$F_{12} = -\tfrac{1}{2}\sqrt{(F_{11}F_{22})} \text{ (empirical, from biaxial test)}$$

- F_i terms (linear) account for the **tension–compression strength difference**; F_{ij} terms are quadratic.
- F_{12} needs a biaxial test; the stability condition is $F_{11}F_{22} - F_{12}^2 > 0$, i.e. $-\sqrt{(F_{11}F_{22})} < F_{12} < \sqrt{(F_{11}F_{22})}$.

- Strength ratio R solves $aR^2 + bR - 1 = 0$: $a = F_{ij}\sigma_i\sigma_j$, $b = F_i\sigma_i \rightarrow R = [-b + \sqrt{(b^2 + 4a)}] / 2a$.
- Shear is **sign-insensitive** ($F_6 = 0$, $F_{16} = F_{26} = 0$ for orthotropic lamina).

Theory	Interaction	Tension ≠ compression	Failure mode	Remark
Max stress	No	Yes	Yes	Simple; unconservative
Max strain	Poisson only	Yes	Yes	Slightly different
Tsai-Hill	Yes	No (select by sign)	No	Smooth; good for UD
Tsai-Wu	Yes	Yes	No	Most general, widely used

VI. Numerical Problems on Strength Theories

Problem 1

Glass/epoxy lamina ($E_1 = 38.6$, $E_2 = 8.27$, $G_{12} = 4.14$ GPa, $\nu_{12} = 0.26$; $X_t = 1062$, $X_c = 610$, $Y_t = 31$, $Y_c = 118$, $S = 72$ MPa) carries $\sigma_1 = 500$, $\sigma_2 = 20$, $\tau_{12} = 30$ MPa. Check failure by all four theories.

(a) Max stress: $500 < 1062$ ✓ ; $20 < 31$ ✓ ; $30 < 72$ ✓ → safe. $R = \min(1062/500, 31/20, 72/30) = \min(2.12, 1.55, 2.40) = \mathbf{1.55}$ (transverse tension governs).

(b) Max strain: $\varepsilon_1 = (500 - 0.26 \times 20)/38600 = 0.01282 < \varepsilon_{1t} = 0.02751$ ✓ ; $\varepsilon_2 = (20 - 0.0557 \times 500)/8270 = -0.00095$ (compressive, limit 0.01427) ✓ ; $\gamma_{12} = 30/4140 = 0.00725 < 0.01739$ ✓ → safe.

(c) Tsai-Hill: $(500/1062)^2 - (500 \times 20)/1062^2 + (20/31)^2 + (30/72)^2 = 0.2217 - 0.0089 + 0.4162 + 0.1736 = \mathbf{0.803 < 1}$ → safe; $R = 1/\sqrt{0.803} = \mathbf{1.12}$.

(d) Tsai-Wu: $F_1 = -6.98 \times 10^{-4}$, $F_2 = 0.02378$, $F_{11} = 1.544 \times 10^{-6}$, $F_{22} = 2.734 \times 10^{-4}$, $F_{66} = 1.929 \times 10^{-4}$, $F_{12} = -1.027 \times 10^{-5}$. Sum $= -0.349 + 0.476 + 0.386 + 0.109 + 0.174 - 0.205 = \mathbf{0.590 < 1}$ → safe; $R = \mathbf{1.34}$.

Observe: Different theories give different margins (1.55 / 1.12 / 1.34); Tsai-Hill is most conservative here because it includes interaction.

Problem 2

The same lamina has fibres at $\theta = 30°$ to the loading axis. Find the maximum σ_x (uniaxial tension) by max stress and by Tsai-Hill.

$\cos^2 30° = 0.75$, $\sin^2 30° = 0.25$, $\sin\theta\cos\theta = 0.433$.

Max stress: $X/\cos^2 = 1062/0.75 = 1416$; $Y/\sin^2 = 31/0.25 = \mathbf{124}$; $S/(sc) = 72/0.433 = 166$ MPa → σ_x = **124 MPa** (matrix failure).

Tsai-Hill: $1/\sigma_x^2 = 0.5625/1062^2 + (1/72^2 - 1/1062^2)(0.1875) + 0.0625/31^2 = 1.0 \times 10^{-4}$ → σ_x = **99.2 MPa** (20 % lower).

VII. Macro Mechanical Analysis of a Laminate

A **laminate** is a stack of laminae with different fibre angles bonded together. **Classical Lamination Theory (CLT)** relates the **in-plane loads N** and **moments M** applied to the laminate to the **mid-plane strains** ε^0 and **curvatures** κ.

7.1 Laminate code (stacking sequence notation)

- Plies listed from the **bottom (or top) to the other face**, angle in degrees, separated by "/": **[0/45/-45/90]**.
- **Subscript s** = symmetric about the mid-plane: $[0/90]_s = [0/90/90/0]$. Subscript **T** = total: $[0/90]_T$ = two plies.
- Subscript number = repeat: $[0/90]_2 = [0/90/0/90]$; $[0/90]_{2s} = [0/90/0/90/90/0/90/0]$.
- **±45** = a +45° ply above a -45° ply ($[\pm 45] = [45/-45]$); an overbar / "o" marks the mid-ply of an odd-ply symmetric laminate: [0/90/0].
- Thicknesses are normally equal, so ply count × t gives total h.

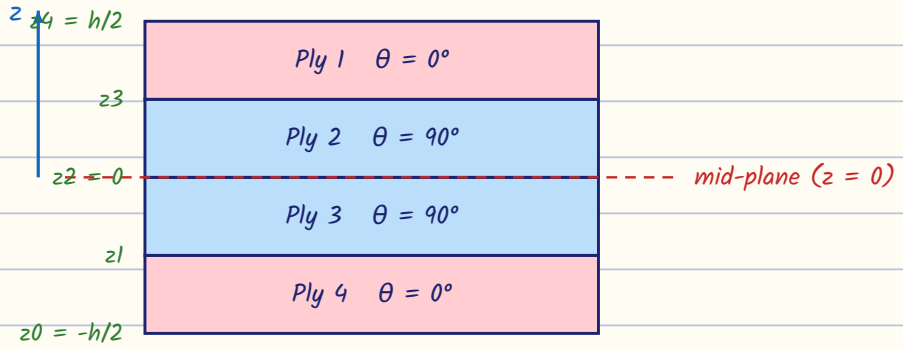


Fig. 2 Laminate [0/90]s with mid-plane and z-coordinates of ply surfaces

Type of laminate	Definition	Example
Symmetric	Plies and angle symmetric about mid-plane: B = 0	[0/45/90]s
Anti-symmetric	Equal ± angles at equal ±z, opposite sign	[45/-45/45/-45]
Balanced	For every +θ ply there is a -θ ply: $A_{16} = A_{26} = 0$	[30/-30]s
Cross-ply	Only 0° and 90° plies	[0/90/0]

Type of laminate	Definition	Example
Angle-ply	Only +θ and -θ plies	[±30]s
Quasi-isotropic	In-plane stiffness same in all directions	[0/±60]s , [0/±45/90]s

VIII. Kirchhoff Hypothesis (Classical Lamination Theory Assumptions)

- The laminate is **thin** ($h \ll$ in-plane dimensions), made of **perfectly bonded** laminae (no slip, bond infinitesimally thin).
- **Normals** to the mid-plane remain **straight** and **normal** to the deformed mid-plane (no transverse shear strains $\gamma_{xz} = \gamma_{yz} = 0$).
- Normals are **inextensible**: the thickness does not change ($\varepsilon_z = 0$).
- Displacements and strains are **small**; each lamina is linear elastic and in **plane stress** ($\sigma_z = \tau_{xz} = \tau_{yz} = 0$).

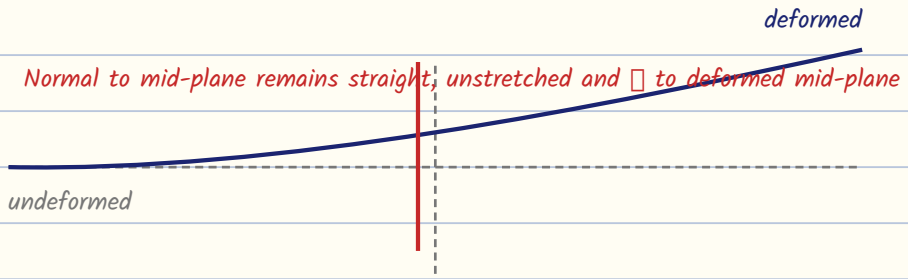


Fig. 3 Kirchhoff-Love kinematics: a normal remains straight and perpendicular after bending

8.1 Displacement and strain field

Let u^0, v^0, w be mid-plane displacements. A point at distance z from the mid-plane moves by:

$$u = u^0 - z\,\partial w/\partial x \;,\; v = v^0 - z\,\partial w/\partial y \;,\; w = w(x, y)$$

Strains from strain-displacement relations ($\varepsilon_x = \partial u/\partial x$, $\varepsilon_y = \partial v/\partial y$, $\gamma_{xy} = \partial u/\partial y + \partial v/\partial x$):

$$\{\varepsilon_x\ \varepsilon_y\ \gamma_{xy}\}^T = \{\varepsilon_x^0\ \varepsilon_y^0\ \gamma_{xy}^0\}^T + z\,\{\kappa_x\ \kappa_y\ \kappa_{xy}\}^T$$

$$\kappa_x = -\partial^2 w/\partial x^2 \;,\; \kappa_y = -\partial^2 w/\partial y^2 \;,\; \kappa_{xy} = -2\,\partial^2 w/\partial x \partial y$$

Strain varies **linearly through thickness** ($\varepsilon = \varepsilon^0 + z\kappa$) - but **stress does not**, since $\bar{Q}$ differs from ply to ply: stress is **piece-wise linear / discontinuous** at ply interfaces.

8.2 Stresses in the k-th ply

$$\{\sigma\}_k = [\bar{Q}]_k\,\{\varepsilon\} = [\bar{Q}]_k\,(\{\varepsilon^0\} + z\,\{\kappa\})$$

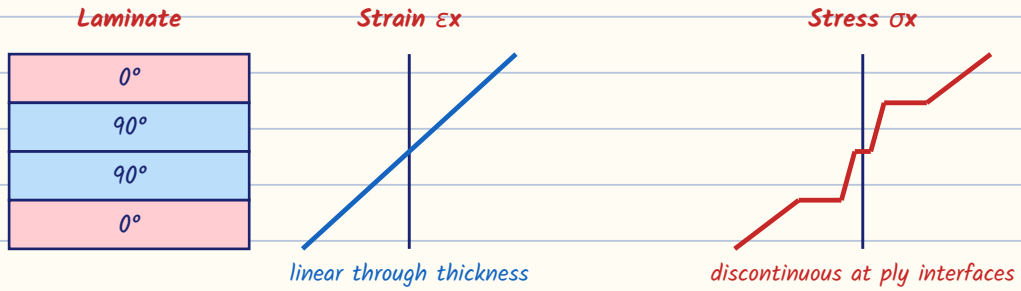


Fig. 4 [0/90]s in bending: linear strain but piece-wise (discontinuous) stress

IX. Force & Moment Resultants – A, B, D Matrices (Derivation)

Replace the stress distribution by **statically equivalent resultants** per unit width, obtained by integrating through the thickness h (z from -h/2 to +h/2):

$$\{N\} = \int \{\sigma\}\, dz \ (N/mm)\ ,\ \{M\} = \int \{\sigma\}\, z\, dz \ (N\cdot mm/mm)$$

$$N = (N_x, N_y, N_{xy})\ ,\ M = (M_x, M_y, M_{xy})$$

Because $\bar{Q}$ is constant within a ply, integrate **ply by ply** between z_{k-1} and z_k (N plies) and substitute $\{\sigma\}_k = [\bar{Q}]_k(\{\varepsilon^0\} + z\{\kappa\})$:

$$\{N\} = \Sigma_{k=1}^{n} \int_{z_{k-1}}^{z_k} [\bar{Q}]_k(\{\varepsilon^0\} + z\{\kappa\})\, dz$$

$$\{M\} = \Sigma_{k=1}^{n} \int_{z_{k-1}}^{z_k} [\bar{Q}]_k(\{\varepsilon^0\}\, z + z^2\{\kappa\})\, dz$$

Since ε^0 and κ are independent of z they come outside the summation; integrating gives:

$$\{N\} = [A]\{\varepsilon^0\} + [B]\{\kappa\}$$

$$\{M\} = [B]\{\varepsilon^0\} + [D]\{\kappa\}$$

$$A_{ij} = \Sigma\, \bar{Q}_{ij,k}\,(z_k - z_{k-1}) \ \text{(extensional stiffness)}$$

$$B_{ij} = \tfrac{1}{2}\,\Sigma\, \bar{Q}_{ij,k}\,(z_k^{\,2} - z_{k-1}^{\,2}) \ \text{(coupling stiffness)}$$

$$D_{ij} = \tfrac{1}{3}\,\Sigma\, \bar{Q}_{ij,k}\,(z_k^{\,3} - z_{k-1}^{\,3}) \ \text{(bending stiffness)}$$

$$\begin{bmatrix} A11 & A12 & A16 & B11 & B12 & B16 \\ A12 & A22 & A26 & B12 & B22 & B26 \\ A16 & A26 & A66 & B16 & B26 & B66 \\ B11 & B12 & B16 & D11 & D12 & D16 \\ B12 & B22 & B26 & D12 & D22 & D26 \\ B16 & B26 & B66 & D16 & D26 & D66 \end{bmatrix}$$

The 6 × 6 laminate stiffness matrix: {N, M} = [[A, B],[B, D]] {, }

Physical meaning

> **[A]** (kN/mm): in-plane (extensional) stiffness - relates N to ε^0. A_{16}, A_{26} = shear-extension coupling.

> **[D]** (kN·mm): bending / twisting stiffness - relates M to κ; D_{16}, D_{26} = bend-twist coupling. $D \propto h^3$ - outer plies dominate.

> **[B]**: **bending-extension coupling**; non-zero only for **unsymmetric** laminates: in-plane load causes bending/warping, and moment causes mid-plane strain.

> Units: A = GPa·mm, B = GPa·mm², D = GPa·mm³.

9.1 Inverted (compliance) form and ply stresses

$$\{\varepsilon^0\} = [a]\{N\} + [b]\{M\}\ ,\ \{\kappa\} = [c]\{N\} + [d]\{M\}$$

[a], [b], [c], [d] from inverting the 6×6 matrix ; for symmetric laminate: $\{\varepsilon^0\} = [A]^{-1}\{N\}$, $\{\kappa\} = [D]^{-1}\{M\}$

Procedure for laminate analysis

1. Get ply properties → [Q] → $[\bar{Q}]_k$ for each ply angle θ_k.
2. Locate z_k from the mid-plane: $z_0 = -h/2$, $z_k = z_{k-1} + t_k$.
3. Compute [A], [B], [D] by the summations above.
4. Solve for $\{\varepsilon^0\}$, $\{\kappa\}$ for given {N}, {M}.
5. Strain in each ply: $\varepsilon = \varepsilon^0 + z\kappa$; stress $\{\sigma\}_{xy} = [\bar{Q}]\{\varepsilon\}$; transform to material axes $\{\sigma\}_{12} = [T]\{\sigma\}_{xy}$.
6. Apply a failure criterion (Tsai-Wu etc.) → **first ply failure (FPF)**, then last ply failure with degraded stiffness.

X. Special Cases of Laminates

Laminate	Effect on [A], [B], [D]	Consequence
Symmetric (mirror about mid-plane)	[B] = 0 ($B_{ij} = 0$ for all i, j)	No bending-stretching coupling; no warping on cure; {N} and {M} decoupled
Balanced (+θ and -θ plies)	$A_{16} = A_{26} = 0$	No shear-extension coupling in plane
Cross-ply (0°/90° only)	$A_{16} = A_{26} = 0$, $D_{16} = D_{26} = 0$, $B_{16} = B_{26} = B_{66} = 0$	Specially orthotropic; unsymmetric cross-ply [0/90] has $B_{11} = -B_{22} \neq 0$ → curved on cooling
Angle-ply $[\pm\theta]_s$	$A_{16} = A_{26} = 0$ but D_{16}, $D_{26} \neq 0$	Bend-twist coupling remains

Laminate	Effect on [A], [B], [D]	Consequence
Quasi-isotropic $[0/\pm60]_s$, $[0/\pm45/90]_s$	$A_{11} = A_{22}$, $A_{66} = (A_{11} - A_{12})/2$, $B = 0$	In-plane isotropic (D is not)
Single isotropic layer	$A = Eh/(1-\nu^2) \cdot [1\ \nu;\ \nu\ 1]$, $B = 0$, $D = Eh^3/12(1-\nu^2)$	Reduces to classical plate theory

Design rule: Use **symmetric & balanced** laminates (e.g. $[0/\pm45/90]_s$) to avoid warping and coupling; place 0° plies outward for bending stiffness.

XI. Numerical Problems on Laminates

Problem 3

Glass/epoxy plies ($E_1 = 38.6$, $E_2 = 8.27$, $G_{12} = 4.14$ GPa, $\nu_{12} = 0.26$), $t = 0.125$ mm each. Find [A], [B], [D] for the laminate $[0/90]_s$.

$Q = [39.17\ 2.18\ 0\ ;\ 2.18\ 8.39\ 0\ ;\ 0\ 0\ 4.14]$ GPa for 0°; for 90°: $Q_{11} = 8.39$, $Q_{22} = 39.17$, $Q_{12} = 2.18$, $Q_{66} = 4.14$. $h = 4 \times 0.125 = 0.5$ mm; $z = -0.25, -0.125, 0, 0.125, 0.25$.

$A_{11} = (2 \times 39.17 + 2 \times 8.39)(0.125) =$ **11.89** ; $A_{22} =$ **11.89** ; $A_{12} = 4 \times 2.18 \times 0.125 =$ **1.09** ; $A_{66} = 4 \times 4.14 \times 0.125 =$ **2.07** (GPa·mm). $A_{16} = A_{26} = 0$.

$[B] =$ **0** (symmetric laminate). $D_{11} =$ **0.368** , $D_{22} =$ **0.1275** , $D_{12} =$ **0.0227** , $D_{66} =$ **0.0431** (GPa·mm³).

Check: $D_{11} \gg D_{22}$ because the 0° plies are at the **outside** (large z) - bending stiffness along x is 3 times that along y.

Problem 4

Same plies. For $[0/90]_s$ under $N_x = 50$ N/mm (others zero), find mid-plane strains, ply stresses and first-ply failure ($Y_t = 31$ MPa).

$[A] = [11.89\ 1.09\ ;\ 1.09\ 11.89]$ kN/mm → $\varepsilon_x^0 = A_{22}N_x/(A_{11}A_{22} - A_{12}^2) =$ **4.24×10^{-3}** ; $\varepsilon_y^0 = -A_{12}N_x/(\dots) =$ **-3.89×10^{-4}** ; effective $E_x = 23.6$ GPa.

0° plies: $\sigma_1 = Q_{11}\varepsilon_x + Q_{12}\varepsilon_y =$ **165.3 MPa** (along fibres, safe vs 1062).

90° plies (material axes): $\sigma_1 =$ **−6.0 MPa**, $\sigma_2 = Q_{12}\varepsilon_y + Q_{22}\varepsilon_x =$ **34.7 MPa > Y_t = 31** → transverse cracking of 90° plies.

FPF load = 50 × 31/34.7 = **44.6 N/mm** (laminate stress 89 MPa); the 0° plies still carry load → last-ply failure is much higher.

Problem 7

Apply Tsai-Wu to the 90° plies of Problem 4 ($\sigma_1 = -6.0$, $\sigma_2 = 34.7$ MPa) to find the strength ratio.

$a = F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + 2F_{12}\sigma_1\sigma_2 = 0.334$; $b = F_1\sigma_1 + F_2\sigma_2 = 0.830$; $a + b = \mathbf{1.165 > 1}$ → fails.

$R = [-b + \sqrt{(b^2 + 4a)}]/2a = [-0.830 + \sqrt{(0.689 + 1.337)}]/0.668 = \mathbf{0.887}$ → FPF at $N_x = 50 \times 0.887 = \mathbf{44.4\ N/mm}$ (agrees with max-stress 44.6).

Problem 5

Find [A], [B] for the unsymmetric cross-ply [0/90] (two plies, t = 0.125 mm).

$A_{11} = (39.17 + 8.39)(0.125) = \mathbf{5.945}$, $A_{12} = \mathbf{0.545}$, $A_{66} = \mathbf{1.035}$ GPa·mm. $B_{11} = ½[39.17(0 - 0.125^2) + 8.39(0.125^2 - 0)] = \mathbf{-0.240}$, $B_{22} = \mathbf{+0.240}$ GPa·mm².

$\mathbf{B \neq 0}$ → a pure in-plane load N_x bends the plate; after cure cooling the plate **warps** (saddle shape).

Problem 6

Angle-ply $[45/-45]_s$, same plies: find A.

$A_{11} = A_{22} = \mathbf{8.56}$; $A_{12} = \mathbf{4.42}$; $A_{66} = \mathbf{5.40}$ GPa·mm ; $A_{16} = A_{26} = 0$ (balanced). B = 0; $D_{16} = D_{26} = \mathbf{0.060} \neq 0$ (bend-twist coupling).

Problem 8

$[0/90]_s$ glass/epoxy (t = 0.125 mm, D from Problem 3) is loaded by $M_x = 1$ N·mm/mm only. Find the curvatures and the extreme-fibre stress.

[D] = [0.3679 0.0227 ; 0.0227 0.1275] kN·mm, $M_x = 0.001$ kN·mm/mm (symmetric: B = 0 → N = 0 ⇒ $\varepsilon^0 = 0$).

det D = 0.3679 × 0.1275 - 0.0227² = 0.04640 ; $\kappa_x = D_{22}M_x/\text{det} = \mathbf{2.748 \times 10^{-3}\ /mm}$, $\kappa_y = -D_{12}M_x/\text{det} = \mathbf{-4.89 \times 10^{-4}\ /mm}$ (anticlastic).

At the outer face z = 0.25 mm: $\varepsilon_x = z\kappa_x = 6.87 \times 10^{-4}$, $\varepsilon_y = -1.22 \times 10^{-4}$ → 0° ply $\sigma_1 = Q_{11}\varepsilon_x + Q_{12}\varepsilon_y = \mathbf{26.6\ MPa}$; at the interface z = 0.125: **13.3 MPa**.

11.1 First-ply failure and last-ply failure

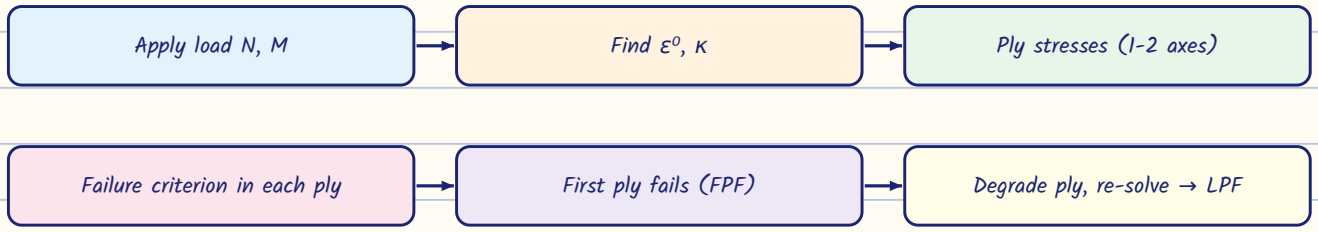


Fig. 5 Progressive failure analysis of a laminate

- **FPF**: load at which the first lamina (usually the 90° / off-axis ply by matrix cracking) satisfies the failure criterion.
- **LPF**: ultimate load after failed plies are discounted (E_2, G_{12}, $\nu_{12} \to 0$) and the load is re-distributed until fibre failure; LPF can be 2-3 times FPF.

XII. Quick Revision - Unit III

Remember

> Max stress / max strain: non-interactive; Tsai-Hill: interactive quadratic; Tsai-Wu: tensor polynomial with F_{12}, handles $X_t \neq X_c$.

> Kirchhoff: normals straight, normal, inextensible → $\varepsilon = \varepsilon^0 + z\kappa$. $N = A\varepsilon^0 + B\kappa$, $M = B\varepsilon^0 + D\kappa$.

> Symmetric → $B = 0$; balanced → $A_{16} = A_{26} = 0$; cross-ply → no coupling terms 16, 26.

Likely exam questions

> Explain max stress, max strain, Tsai-Hill and Tsai-Wu theories; compare them.

> State Kirchhoff hypothesis; derive A, B, D matrices in detail.

> Explain special cases of laminates; numerical on A, B, D and first-ply failure.

Short Answer Questions (2 marks)

Q1. List the five basic strengths of a lamina.

Ans: X_t, X_c, Y_t, Y_c, S.

Q2. Which theory is non-interactive?

Ans: Maximum stress and maximum strain theories.

Q3. Write Tsai-Hill criterion for plane stress.

Ans: $\sigma_1^2/X^2 - \sigma_1\sigma_2/X^2 + \sigma_2^2/Y^2 + \tau_{12}^2/S^2 = 1$.

Q4. Why is the F_{12} term required in Tsai-Wu?

Ans: It accounts for interaction between σ_1 and σ_2 (biaxial); it must be found from a biaxial test or approximated as $-\frac{1}{2}\sqrt{(F_{11}F_{22})}$.

Q5. State Kirchhoff's hypothesis.

Ans: Normals to the mid-plane stay straight, normal to the deformed mid-plane and unchanged in length.

Q6. What does [B] represent?

Ans: Bending-extension coupling stiffness; zero for symmetric laminates.

Q7. When are A_{16} and A_{26} zero?

Ans: For balanced laminates (equal $+\theta$ and $-\theta$ plies) and cross-ply laminates.

Q8. What is a quasi-isotropic laminate?

Ans: A laminate (e.g. $[0/\pm 60]_s$) whose in-plane stiffness is the same in all directions.

Q9. Define first-ply failure.

Ans: Load at which the first ply of the laminate satisfies a failure criterion.

Practice Problems with Answers

Problem A

Glass/epoxy lamina: $\sigma_1 = 300$, $\sigma_2 = -50$, $\tau_{12} = 40$ MPa. Use Tsai-Hill with strengths $X_t = 1062$, $Y_c = 118$, $S = 72$.

$(300/1062)^2 - 300(-50)/1062^2 + (50/118)^2 + (40/72)^2 = 0.0798 + 0.0133 + 0.1796 + 0.3086 = 0.581 < 1$ → safe, $R = 1.31$.

Problem B

Check max-stress failure of a lamina with $\sigma_1 = 1200$, $\sigma_2 = 10$, $\tau_{12} = 20$ MPa (same material).

$\sigma_1 = 1200 > X_t = 1062$ → **fibre failure** ($R = 0.885$).

UNIT IV

Metal Matrix Composites (MMCs)

Reinforcements & base metals | need & applications | fabrication processes | properties

I. Introduction to Metal Matrix Composites

A **Metal Matrix Composite (MMC)** consists of a **metal or alloy matrix** (Al, Mg, Ti, Cu, Ni ...) reinforced by a **hard, stiff, strong second phase** (ceramic fibres, whiskers or particles; sometimes a second metal). The metal gives ductility, toughness and conductivity; the reinforcement gives stiffness, strength, wear and creep resistance and low thermal expansion.

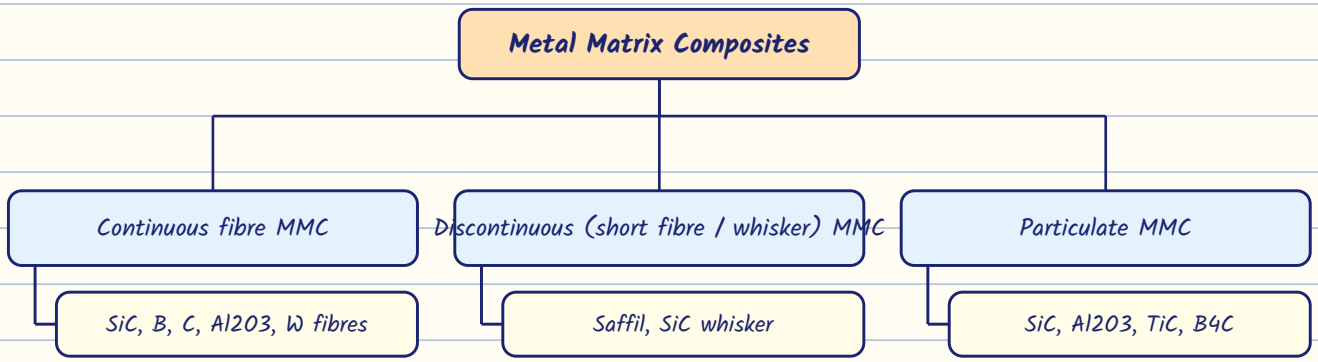


Fig. 1 Classification of MMCs by reinforcement type

I.1 Advantages over monolithic metals and polymer matrix composites

- **High specific strength and stiffness**, tailored thermal expansion (CTE) - matches Si, GaAs and ceramics.
- **Higher service temperature** than PMCs (Al: 300 °C, Ti: 600 °C, Ni: > 1000 °C) and retention of properties at heat.
- Excellent **wear and abrasion resistance**, fatigue and creep resistance; high thermal and electrical conductivity.
- **No moisture absorption, no out-gassing**, non-flammable, resistant to radiation and UV; better transverse and shear strength than PMCs.
- Can be machined, joined, heat-treated by metal-working methods (extrusion, forging, rolling).

I.2 Limitations

- **High cost** of reinforcements and processing; heavier than PMCs ($\rho \approx 2.8$-4.5 g/cc).

- **Reinforcement – matrix reaction** at high temperature (Al_4C_3 in C/Al; brittle phases at interface); poor wetting of ceramics by molten metal.
- Reduced ductility, fracture toughness; **difficult to machine** (tool wear) and to recycle; galvanic corrosion at the interface.

II. Reinforcement Materials

Reinforcements are selected for high strength / modulus, low density, **chemical compatibility** and **wettability** with the matrix, thermal stability, appropriate CTE and cost.

2.1 Types and characteristics

Type	Examples	Typical properties	Remark
Continuous fibres	Carbon (graphite), boron, SiC (SCS-6, Nicalon), Al_2O_3 (Nextel, FP), tungsten, steel wire	Diameter 8–140 µm; E = 200–600 GPa; σ = 2–4 GPa	Highest, anisotropic properties; costly; need coating
Short fibres / whiskers	Saffil (δ-Al_2O_3), SiC whiskers, K-titanate, Kaowool	d = 0.1–10 µm, L/d = 10–100; E = 300–500 GPa	Near isotropic; health hazard of whiskers
Particulates	SiC, Al_2O_3, B_4C, TiC, TiB_2, WC, graphite, fly ash, Si_3N_4	Size 1–100 µm; hardness 20–35 GPa	Cheapest, isotropic, easy casting and PM; most widely used
Metal wire	Mo, W, steel, Nb	High density, very ductile	Used in Cu, Ni, Al matrices

Reinforcement	Density (g/cc)	E (GPa)	CTE (10^{-6}/K)	Strength (MPa)
SiC	3.2	400–480	4.3	2000–3000 (fibre / whisker)
Al_2O_3	3.9	350–400	7–8	1500–2000
B_4C	2.5	450	5.0	~ 3000
Carbon (HM)	1.9–2.2	400–600	−1 to 0	2500–3500
Boron	2.6	400	5.0	3500
TiC	4.9	440	7.4	> 1500

2.2 Selection criteria for reinforcement

- **High modulus & strength**, low density → good specific properties.
- **Chemically stable** with the matrix at fabrication and service temperature (reaction layer must be thin, < 100 nm).
- **Wettability** by molten metal (Mg or Ti additions improve wetting of SiC / Al_2O_3 by Al); coatings (SiC, TiB_2, C, Ni) on fibres.
- **CTE mismatch** small to avoid residual stresses and thermal fatigue; **cost** and availability; safety (whisker toxicity).

III. Base Metals (Matrix) – Types and Selection

Matrix	Density (g/cc)	Max. use temp.	Reinforcements used	Applications
Aluminium (6061, 2124, A356)	2.7	≈ 300 °C	SiC, Al_2O_3, B_4C, C, B	Pistons, brake discs, aerospace struts, electronics
Magnesium (AZ91, ZK60)	1.74	≈ 250 °C	SiC, C, Al_2O_3	Lightest MMC: aerospace, auto
Titanium (Ti-6Al-4V)	4.5	≈ 600 °C	SiC fibre, TiC, TiB, B_4C	Jet-engine, aerospace skin
Copper	8.9	≈ 400 °C	Graphite, W, Mo, SiC, Al_2O_3	Heat sinks, electrical contacts
Nickel / superalloys	8.2–8.9	> 1000 °C	W, SiC, Al_2O_3, TaC	Turbine blades, hot sections
Iron / steel	7.8	≈ 600 °C	TiC, WC, Al_2O_3	Wear tools, dies
Zinc, Lead, Tin	7.1 / 11.3 / 7.3	low	Graphite, SiC	Bearings, low-temperature

Criteria for selecting the base metal

> **Service temperature** and environment (oxidation, corrosion).
> **Density / specific strength** required (Mg < Al < Ti < Cu < Ni).
> **Compatibility and wetting** with reinforcement; reaction products.
> **Processability**: castability, formability, joinability, heat-treatability (Al 6061-T6, 2124).
> **Cost and availability**; conductivity requirement (Cu, Al for thermal management).

IV. Need for Production of MMCs and Applications

4.1 Why MMCs are needed

- Monolithic alloys cannot give **high specific stiffness, wear resistance, low CTE and high-temperature strength together**.
- Demand for **lightweight** fuel-efficient vehicles and aircraft (weight saving 20-40 %).
- **Thermal management** of electronics: heat must be removed while CTE matches the chip (Al-SiC, Cu-W).
- Replace **cast iron** in brakes and cylinder liners (lighter, better wear), and expensive Ti / Ni in some hot parts.
- Better **creep, fatigue and dimensional stability** for space structures.

4.2 Applications

Sector	Components (MMC used)
Automobile	Pistons, piston crowns, cylinder liners, connecting rods, brake rotors / drums (Al-SiC), drive shaft (Al-SiC), engine blocks, calipers
Aerospace & defence	Space-shuttle orbiter mid-fuselage struts (B/Al), satellite structures (Gr/Mg), fan-exit guide vanes, rotor blades, missile fins, helicopter blade sleeves, armour
Electronics & thermal	Microprocessor heat sinks / substrates, IGBT base plates, RF packages (Al-SiC, Cu-W, Cu-Mo, Al-Gr)
Sports & recreation	Bicycle frames, golf clubs, tennis rackets (Al-Al_2O_3)
Industrial	Cutting tools (WC-Co), dies, pump parts, rollers, bearings, wear plates

V. Fabrication Processes for MMCs

MMC production methods are grouped as **solid-state (powder metallurgy, diffusion bonding)**, **liquid-state (stir casting, infiltration, squeeze casting, spray deposition)**, **vapour-state** and **in-situ** processes.

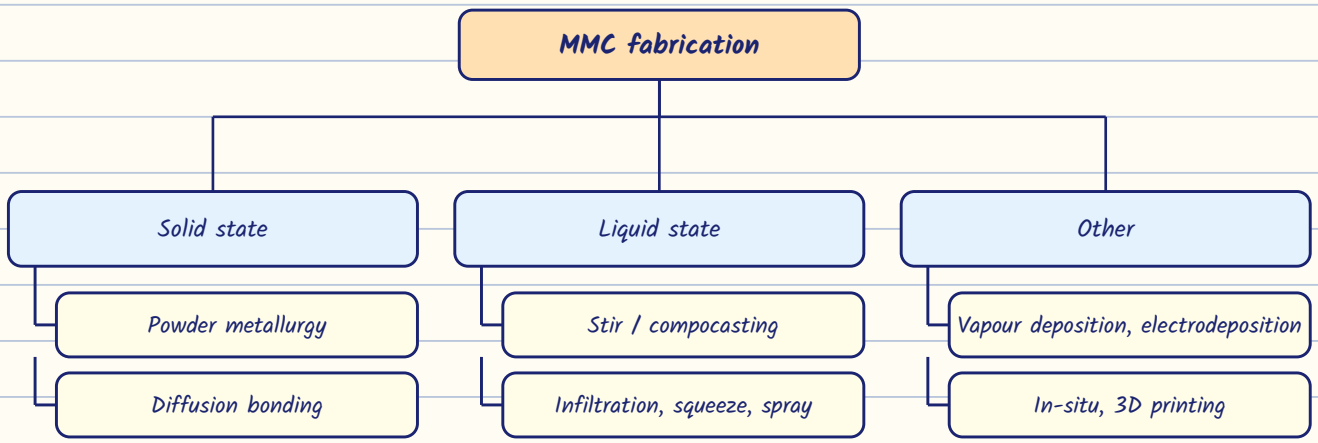


Fig. 2 Routes for manufacturing MMCs

5.1 Powder metallurgy (PM) technique

Metal powder (10-50 μm) and reinforcement (particles / short fibres) are **blended**, **compacted** and **consolidated** below the melting point of the matrix - no melt, so no reaction, and high V_f (up to 50 %) is possible.

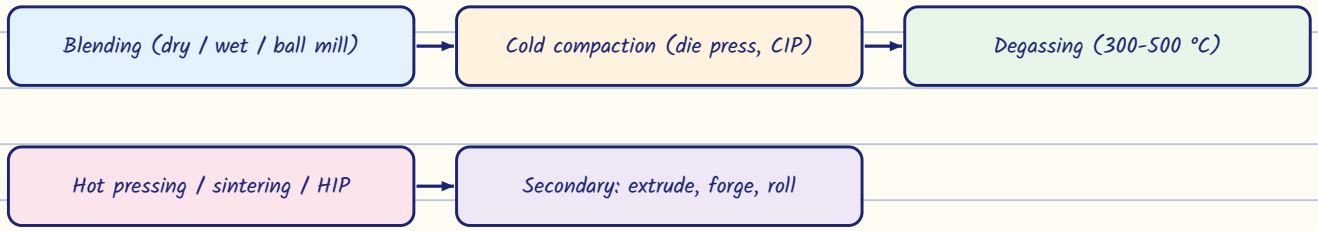


Fig. 3 Powder metallurgy route

- **Blending**: uniform mixing; ball-milling / mechanical alloying breaks agglomerates.
- **Cold pressing** (100-400 MPa) gives a "green" compact; **degassing** removes moisture and oxides.
- **Consolidation**: hot pressing, vacuum sintering, hot isostatic pressing (HIP) or extrusion to near full density.
- **Advantages**: fine microstructure, high V_f, no chemical reaction, near-net shape. **Limitations**: costly powder, contamination (oxide films), size limit, residual porosity, safety (fine Al powder).

5.2 Liquid metallurgy techniques

Reinforcement is put into **molten** matrix; cheapest and most scalable route (casting).

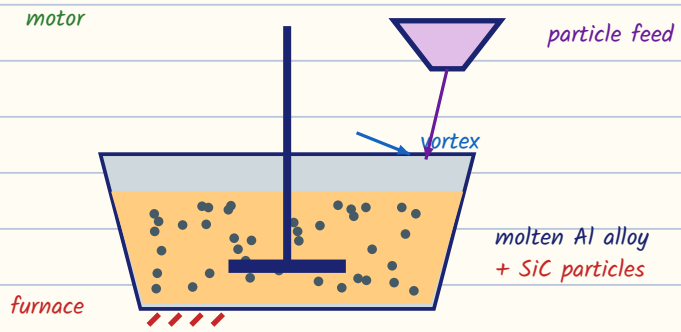


Fig. 4 Stir casting (vortex method)

- **Stir casting (vortex)**: preheated SiC (≈ 900 °C) is fed into a **vortex** created by a mechanical stirrer in molten Al (≈ 700 °C) → cast in a mould. Mg is added to improve wettability. Up to 30 % V_p; problems: particle settling, clustering, porosity (argon degassing).
- **Compocasting / rheocasting**: particles are added to **semi-solid (partially solidified) slurry**: viscosity holds particles in suspension, reaction is minimal.
- **Squeeze casting / pressure infiltration**: molten metal forced by a **hydraulic punch (50–150 MPa)** into a **preform** of fibres / particles; no porosity, high V_f (up to 60 %), near-net shape.

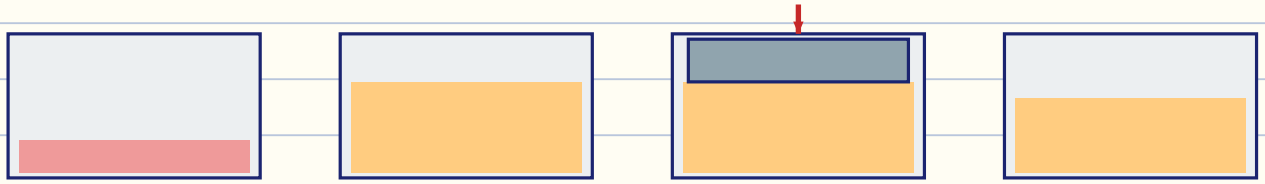


Fig. 5 Squeeze-casting infiltration

- **Gas-pressure infiltration**: argon pressure forces the melt into a preform (large parts, e.g. Al-SiC substrates); **vacuum infiltration** relies on capillarity + vacuum.
- **Spray deposition (Osprey process)**: molten alloy is **atomised** by inert gas, and reinforcement particles are injected into the spray; droplets + particles deposit on a substrate as a near-dense preform; rapid solidification gives fine grains.
- **In-situ (reactive) processing**: the reinforcement is **formed in the melt** by chemical reaction (e.g. Al + K_2TiF_6 + KBF_4 → TiB_2 particles; XD™ process); clean, well-bonded interface.

5.3 Secondary processing

- **Extrusion, rolling, forging, drawing** at 400–500 °C: break clusters, align whiskers / short fibres, close pores, improve strength and ductility.

- **Heat treatment** (solution + ageing, T6): precipitation hardening of the Al matrix is accelerated by dislocations around particles.
- **Machining, joining**: diamond / PCD tools, EDM, laser, abrasive water-jet; friction-stir / diffusion welding.
- **Superplastic forming** of fine-grained MMC sheets; HIP for densification of castings.

5.4 Special fabrication techniques

Technique	Principle	Used for
Diffusion bonding	Alternate layers of matrix foil and fibre mats (or coated fibres) are stacked and pressed 20-100 MPa at 0.6-0.8 T_m for hours; atoms diffuse across the interfaces (foil-fibre-foil)	Ti / SiC, Al / B sheets, plates
Vapour deposition (PVD / CVD / plasma spray)	Matrix metal is vapour-deposited or plasma-sprayed on to fibres (monotape), then hot-pressed	Fibre MMC tapes, turbine parts
Electrodeposition / electroforming	Co-deposition of particles / fibres while metal (Ni, Cu) is plated from the bath	Coatings, Ni-SiC wear surfaces
Mechanical alloying	High-energy ball milling repeatedly fractures and cold-welds powders → nano-dispersion	ODS alloys, nano MMC
Additive manufacturing / laser cladding	Powder or wire mixed with reinforcement fused layer by layer	Complex, graded MMCs
Friction stir processing	Rotating tool stirs reinforcement into a solid-state surface	Surface composites

Process	V_f	Cost	Reaction risk	Product
Stir casting	≤ 30 %	Low	Moderate	Large castings
Squeeze casting	≤ 60 %	Medium	Low	Near-net parts
Powder metallurgy	≤ 50 %	High	Very low	Small, high quality
Spray deposition	≤ 20 %	Medium-high	Low	Billets
Diffusion bonding	≤ 80 %	Very high	Low-medium	Fibre sheets

VI. Properties of MMCs

6.1 Physical properties

- **Density** by rule of mixtures: $\rho_c = \rho_m V_m + \rho_r V_r$ - e.g. Al + 20 % SiC → 2.8 g/cc (4 % heavier than Al).

- **Coefficient of thermal expansion (CTE)** decreases with V_r (Turner / Kerner models) and can be tailored: Al-SiC 55-65 % → 6.5-8 $\times 10^{-6}$/K ≈ chip & ceramic substrates.
- **Thermal conductivity** is high (Al-SiC 170-200 W/mK, Cu-graphite up to 500); electrical conductivity decreases with V_r.
- **Damping capacity** is higher than monolithic metal (interface, dislocation damping).

6.2 Mechanical properties

Upper bound: $E_c = E_m V_m + E_r V_r$ **Lower bound:** $1/E_c = V_m/E_m + V_r/E_r$

- **Modulus, yield & tensile strength increase** with V_r (Al-20 % SiC: E ≈ 100-115 GPa vs 70 GPa for Al; σ_y up 30-60 %); strength mechanisms: load transfer, Orowan strengthening, grain refinement (Hall-Petch), dislocation density from CTE mismatch, work hardening.
- **Ductility and fracture toughness drop** (elongation from 15 % to 1-5 %; K_{Ic} 40 → 15-25 MPa√m).
- **Fatigue strength** improves (crack initiation delayed) but crack growth is faster at high ΔK; **creep resistance** much better (particles pin dislocations, grain boundaries).
- **Stiffness in fibre direction**: continuous B/Al: E_l = 220 GPa, σ_l = 1.4 GPa at V_f = 0.5.

6.3 Wear properties

- Hard ceramic particles **carry the load and resist abrasion** - wear rate falls with V_r (Al-20 % SiC wears about 5-10 times slower than Al alloy).
- Adhesive wear and **seizure** are reduced; graphite particles act as a **solid lubricant** (Al-Gr bearings, cylinder liners).
- Mechanically mixed layer (MML) of oxides and debris forms on the worn surface and protects the sliding surface; large, hard particles are better against abrasion, small particles for adhesion.
- Counter-face wear can increase (hard particles abrade the mating steel) - use compatible materials.

6.4 Machinability

- **Poor machinability**: hard abrasive ceramic particles wear HSS / carbide tools rapidly (flank wear); built-up edge is smaller, surface finish rough due to particle pull-out.
- Use **PCD (polycrystalline diamond)** or CVD diamond tools, low feed, high speed (≈ 300-600 m/min), liquid coolant.
- Non-conventional methods: **EDM**, wire-EDM, laser, abrasive water-jet and ultrasonic machining; near-net-shape casting to minimise machining.

- Wear depends on size and V_r of the particle: tool life falls by 10-50 times for 20-30 % SiC.

6.5 Other properties

- **Corrosion**: galvanic coupling between C / SiC and Al can raise pitting; surface anodising and sealing needed.
- **High-temperature stability**: Al-SiC useful to 300 °C; Ti-SiCf to 650 °C. **Dimensional stability** and low outgassing for space.
- Good **electromagnetic shielding** and **non-magnetic** behaviour (Al, Mg MMCs).

VII. Effect of Size, Shape and Distribution of Particulate on Properties

Factor	Effect on properties
Particle size	Smaller particles (< 10 µm) → higher yield and tensile strength (Orowan / Hall-Petch, more interfaces), better ductility and toughness; large particles (> 30 µm) crack easily and promote void nucleation, but give better wear / abrasion resistance
Volume fraction	Modulus, strength, hardness, wear resistance and thermal stability ↑; ductility and toughness ↓; castability and machinability worsen above ≈ 25-30 %
Shape	Angular / sharp particles → high local stress concentration, lower ductility, but stronger interlocking. Spherical → better flow, higher ductility. High aspect ratio (whiskers, fibres) → much higher strength / modulus (load transfer) but anisotropic
Distribution	Uniform → best properties. **Clustering / agglomeration** and particle-free zones → premature failure, porosity, scatter and low ductility. Segregation by settling in casting (density difference), lessened by stirring, compocasting, extrusion
Interface	Good bonding → load transfer; excessive reaction layer (Al_4C_3) → brittle; weak interface → debonding

Concept - particle strengthening

> Orowan: $\Delta\sigma_y = 0.13\ G\ b / \lambda \cdot \ln(r/b)$ → **smaller spacing λ = stronger**.

> Shear-lag (load transfer): $\Delta\sigma = \sigma_m\ V_r\ (S + 4)/4$ for aspect ratio $S = L/d$.

> CTE mismatch: $\Delta\alpha\ \Delta T$ creates dislocations → more strengthening; also residual stresses.

VIII. Numerical Problems on MMCs

Problem 1

Al-20 vol % SiC particulate MMC: $\rho_{Al} = 2.7$, $\rho_{SiC} = 3.2$ g/cc, $E_{Al} = 70$, $E_{SiC} = 450$ GPa. Find ρ_c and the upper / lower bounds of E_c.

ρ_c = 2.7(0.8) + 3.2(0.2) = **2.80 g/cc**

Upper: E_c = 70(0.8) + 450(0.2) = 56 + 90 = **146 GPa**

Lower: $1/E_c$ = 0.8/70 + 0.2/450 = 0.011429 + 0.000444 → E_c = **84.2 GPa** (test value ≈ 100–115 GPa lies between the bounds).

Specific modulus: Al 70/2.7 = 26 → MMC ≈ 105/2.8 = 37.5 GPa/(g/cc): **40 % better**.

Problem 2

Aluminium-boron fibre composite with V_f = 0.5: E_f = 400 GPa, E_m = 70 GPa, σ_{fu} = 3500 MPa. Estimate E_1, E_2 and the strength along fibres (neglect matrix contribution).

E_1 = 400(0.5) + 70(0.5) = **235 GPa**. E_2 = 400 × 70 / (400 × 0.5 + 70 × 0.5) = 28000 / 235 = **119 GPa**.

$\sigma_{lu} \approx \sigma_{fu} V_f$ = 3500 × 0.5 = **1750 MPa** (+ small matrix part) → compare Al alloy ≈ 300 MPa.

IX. Quick Revision – Unit IV

Remember

- MMC = metal matrix + ceramic reinforcement: higher T, wear, conductivity, no moisture than PMC; costly, less ductile.
- Reinforcement: SiC, Al_2O_3, B_4C, C, B, W; matrix: Al, Mg, Ti, Cu, Ni, Fe.
- Fabrication: **PM** (blend-press-sinter-HIP), **stir casting**, **compocasting**, **squeeze / pressure infiltration**, **spray deposition**, in-situ, diffusion bonding, vapour deposition; secondary: extrusion, rolling, heat-treat.
- Properties: ↑ E, σ, wear, creep; ↓ ductility, toughness, machinability. Small, uniform, well-bonded particles are best.

Likely exam questions

- Classify MMCs. Discuss reinforcement materials and selection of base metals.
- Explain need for MMCs and applications.
- Describe powder metallurgy and liquid metallurgy techniques; write about secondary and special fabrication processes.
- Write a note on physical, mechanical, wear and machining properties. Effect of size, shape and distribution of particulates.

Short Answer Questions (2 marks)

Q1. What is an MMC?

Ans: Composite with a metal or alloy matrix reinforced with ceramic / metallic fibres, whiskers or particles.

Q2. Give two advantages of MMCs over PMCs.

Ans: Higher service temperature, no moisture absorption, better wear and thermal / electrical conductivity.

Q3. Which reinforcement is most common for Al MMCs?

Ans: SiC and Al_2O_3 particles.

Q4. Why is Mg added in stir casting of Al-SiC?

Ans: Mg lowers the surface tension and improves wetting of SiC by molten Al.

Q5. What is compocasting?

Ans: Stir casting in the semi-solid state so that particles are held in suspension.

Q6. What is squeeze casting?

Ans: Forcing molten metal into a preform under high pressure and solidifying under that pressure.

Q7. Purpose of degassing in powder metallurgy?

Ans: Removal of adsorbed moisture and gases before consolidation.

Q8. Name machining tools for MMC.

Ans: PCD / diamond-coated tools; EDM, laser, abrasive water jet.

Q9. How does particle size affect strength?

Ans: Smaller particles give higher strength and ductility; large particles crack and reduce toughness.

Q10. Which MMC is used for electronic packaging?

Ans: Al-SiC, Cu-W, Cu-Mo.

Glossary of Key Terms

Term	Meaning
Wettability	Ability of molten metal to spread on a ceramic surface
Interface	Boundary region between matrix and reinforcement where load is transferred
Preform	Porous skeleton of fibres / particles into which metal is infiltrated
HIP	Hot isostatic pressing: high temperature + gas pressure from all sides
ODS	Oxide dispersion strengthened alloy
CTE	Coefficient of thermal expansion
Orowan strengthening	Strengthening by dislocations bowing between hard particles
Whisker	Single-crystal short fibre of very high strength

Answer Outlines for 10-mark Questions

Powder metallurgy of MMC

> Blending of powders, sketch of steps.
> Cold compaction, degassing, sintering / hot pressing / HIP, extrusion.
> Advantages (fine structure, high V_f, no reaction) and limitations.

Stir casting and squeeze casting

> Sketch of stir casting set-up, role of vortex and Mg.
> Problems: settling, clustering, porosity, reaction.
> Squeeze casting procedure, pressure range, advantages.

Effect of particle size, shape and distribution

> Size: Orowan, Hall-Petch; ductility trend.
> Shape: angular vs spherical; whiskers (aspect ratio).
> Distribution: clusters, particle free zones; remedy by secondary working.

Practice Problems with Answers

Problem A

Cu-W composite with 60 vol % W (ρ_{Cu} = 8.9, ρ_W = 19.3 g/cc). Find density.

$\rho_c = 8.9(0.4) + 19.3(0.6) = 3.56 + 11.58 =$ **15.14 g/cc**.

Problem B

Al-30 vol % SiC (E_{Al} = 70, E_{SiC} = 450 GPa): find the upper bound modulus and the weight fraction of SiC (ρ = 2.7 / 3.2).

$E_c = 70(0.7) + 450(0.3) =$ **184 GPa** ; $\rho_c = 2.7(0.7) + 3.2(0.3) = 2.85$; $W_{SiC} = 3.2(0.3)/2.85 =$ **0.337**.

UNIT V

Failure Theories of Composites

Micromechanics of failure | anisotropic strength theories | importance of shear strength | choice of criteria

I. Micromechanics of Failure of a Unidirectional Lamina

The strength of a lamina depends on **how the fibre, matrix and interface fail**. Micromechanics of failure explains the five basic strengths (X_t, X_c, Y_t, Y_c, S) from constituent properties and identifies the **failure mode** for each loading.

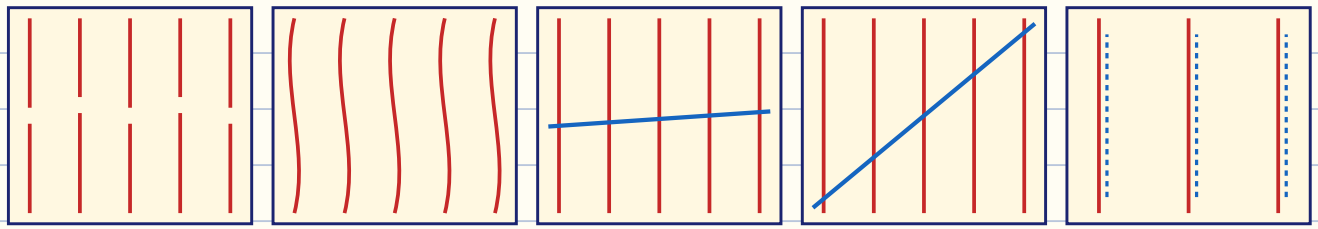


Fig. 1 Typical failure modes of a unidirectional lamina

Loading	Dominant failure mechanism	Controlled by
Longitudinal tension (X_t)	Fibre fracture (+ pull-out, matrix cracking, debonding)	Fibre strength, V_f
Longitudinal compression (X_c)	Fibre microbuckling / kink band, shear crushing	Matrix shear modulus, fibre alignment
Transverse tension (Y_t)	Matrix cracking, interface debonding, voids	Matrix & interface strength
Transverse compression (Y_c)	Matrix shear failure on 45° plane	Matrix strength
In-plane shear (S)	Matrix shear cracking, fibre-matrix interface shear	Matrix, interface

I.1 Longitudinal tensile strength Xt

Fibres are **brittle with scatter in strength** (flaws), so they break at random points at different loads. After a fibre breaks, load is transferred back to it through the matrix over an **ineffective length**, and neighbouring fibres see a stress concentration (Rosen's chain-of-bundles / Zweben models). Two cases decide the simple strength estimate ($\varepsilon_{fu} < \varepsilon_{mu}$, brittle fibre):

$$\sigma_{lu} = \sigma_{fu} V_f + \sigma'_m (1 - V_f) \text{ for } V_f > V_{crit}$$
$$\sigma_{lu} = \sigma_{mu}(1 - V_f) \text{ for } V_f < V_{min} \text{ (fibres only dilute the matrix)}$$
$$V_{min} = (\sigma_{mu} - \sigma'_m) / (\sigma_{fu} + \sigma_{mu} - \sigma'_m), \quad V_{crit} = (\sigma_{mu} - \sigma'_m) / (\sigma_{fu} - \sigma'_m)$$

σ_{fu} = fibre ultimate strength, σ_{mu} = matrix ultimate strength, σ'_m = matrix stress when the fibres fail ($\sigma'_m = E_m \varepsilon_{fu}$). In practice $X_t \approx 0.8$-0.9 of the rule of mixtures because of fibre misalignment and stress concentration.

Example

> $\sigma_{fu} = 2000$ MPa, $\sigma_{mu} = 60$ MPa, $\sigma'_m = 30$ MPa → $V_{min} = 30/2030 =$ **1.5 %**, $V_{crit} = 30/1970 =$ **1.5 %** : only above ≈ 1.5 % fibre the fibres begin to strengthen the matrix.

> At $V_f = 0.6$: $\sigma_{lu} = 2000(0.6) + 30(0.4) =$ **1212 MPa**.

1.2 Longitudinal compressive strength Xc

Under compression the fibres behave like columns on an elastic foundation (the matrix) and **microbuckle**. Rosen's analysis gives two modes:

Extension (in-phase) mode: $\sigma_c = 2V_f \sqrt{(V_f E_f E_m / 3(1 - V_f))}$

Shear (out-of-phase) mode: $\sigma_c = G_m / (1 - V_f)$

Example (glass/epoxy, $V_f = 0.6$, $E_f = 72$, $E_m = 3.5$, $G_m = 1.3$ GPa): extension mode = $2(0.6)\sqrt{(0.6 \times 72 \times 3.5/1.2)} =$ **13.5 GPa**; shear mode = $1.3/0.4 =$ **3.25 GPa** → the **shear mode governs**. Actual X_c (≈ 0.6-1.2 GPa) is far lower because of **fibre misalignment**, which gives kink-band formation at $\sigma_c \approx \tau_y/\varphi$ (φ = initial misalignment angle). Hence $X_c < X_t$ especially for aramid fibres.

1.3 Transverse tensile strength Yt

- Fibres are stiffer than the matrix, so the **matrix strain is magnified** between fibres: $\varepsilon_m = S_m \varepsilon_2$, where strain-magnification factor $S_m = 1 / [1 - V_f(1 - E_m/E_f)]$ (≈ 2.5 for $V_f = 0.6$ glass/epoxy).
- Failure starts at **interface debonding**, voids or matrix cracking: $Y_t \approx \sigma_{mu}/S_m$ or lower; so Y_t is only 3-5 % of X_t and is the **weakest** strength - the first failure in a laminate.
- Improved by good fibre-matrix adhesion (sizing), tougher matrix, lower void content.

1.4 Transverse compressive strength Yc

- Matrix compression crushes the matrix: failure is by **shear on planes at ≈ 45°-55°** to the load; fibres reinforce the matrix so $Y_c \approx 3$-$5 \times Y_t$.
- Interface crushing and fibre-matrix debonding add to the failure.

1.5 In-plane shear strength S

- Shear strain is carried mainly by the matrix: **strain magnification** ($\gamma_m = \gamma_{12}/[1 - V_f(1 - G_m/G_f)]$) leads to matrix yielding / cracking at 45° to the fibres.
- S is of the same order as $Y_t \cdot 2$ (typ. 60-100 MPa) and is strongly non-linear (shear stress-strain curve softens).

II. Anisotropic Strength and Failure Theories

Because composite strength is **directional**, isotropic criteria (Tresca, von Mises) fail. Anisotropic criteria use the **five basic strengths** and are applied in the **material axes**.

2.1 Variation of strength with fibre angle

A lamina loaded in tension at angle θ to the fibres has strength $\sigma_x(\theta)$:

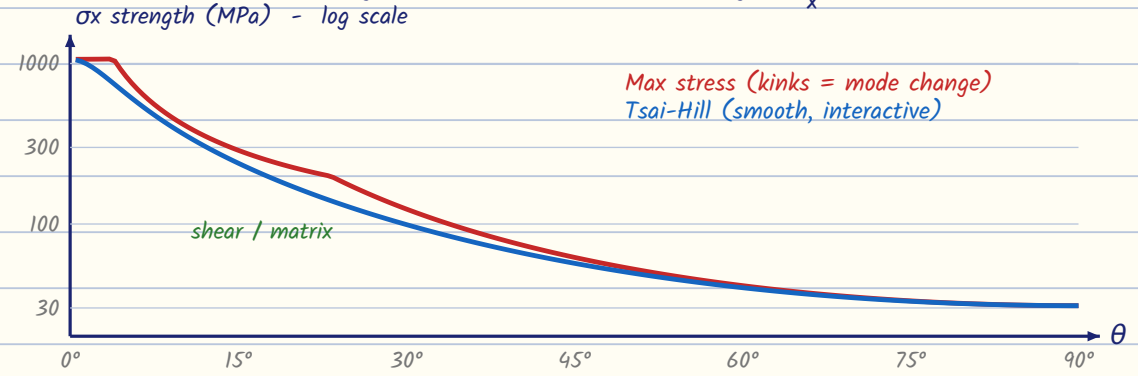


Fig. 2 Off-axis tensile strength of glass/epoxy vs fibre angle

- A small misalignment (≈ 5-10°) cuts the strength by **60-65 %**: shear and transverse stresses appear immediately (at 10°: $\sigma_1 = 0.97\sigma_x$, $\sigma_2 = 0.03\sigma_x$, $\tau = 0.17\sigma_x$) and the weak S and Y are exceeded.
- Near 0° → fibre-controlled; 10°-40° → **shear-controlled**; near 90° → transverse (matrix) controlled.
- The **max stress** theory gives a jagged curve; **Tsai-Hill** / **Tsai-Wu** give a smooth curve that matches test data (interaction).

2.2 Summary of criteria

Criterion	Equation (plane stress)
Max stress	$\sigma_1 = X$, $\sigma_2 = Y$, $\lvert\tau_{12}\rvert = S$ (any one)
Max strain	$\varepsilon_1 = X/E_1$, $\varepsilon_2 = Y/E_2$, $\gamma_{12} = S/G_{12}$

Criterion	Equation (plane stress)
Hill / Tsai-Hill	$\sigma_1^2/X^2 - \sigma_1\sigma_2/X^2 + \sigma_2^2/Y^2 + \tau_{12}^2/S^2 = 1$
Hoffman	$\sigma_1^2/(X_tX_c) - \sigma_1\sigma_2/(X_tX_c) + \sigma_2^2/(Y_tY_c) + (X_c - X_t)\sigma_1/(X_tX_c) + (Y_c - Y_t)\sigma_2/(Y_tY_c) + \tau_{12}^2/S^2 = 1$
Tsai-Wu	$F_1\sigma_1 + F_2\sigma_2 + F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + F_{66}\tau_{12}^2 + 2F_{12}\sigma_1\sigma_2 = 1$
Azzi-Tsai	Tsai-Hill with $\sigma_1\sigma_2$ term replaced by $\lvert\sigma_1\sigma_2\rvert$ (equal tension/compression strengths); used for ±θ laminates
Hashin (mode-based)	Fibre tension: $(\sigma_1/X_t)^2 + (\tau/S)^2 = 1$; matrix tension: $(\sigma_2/Y_t)^2 + (\tau/S)^2 = 1$; fibre compression: $\sigma_1 = -X_c$

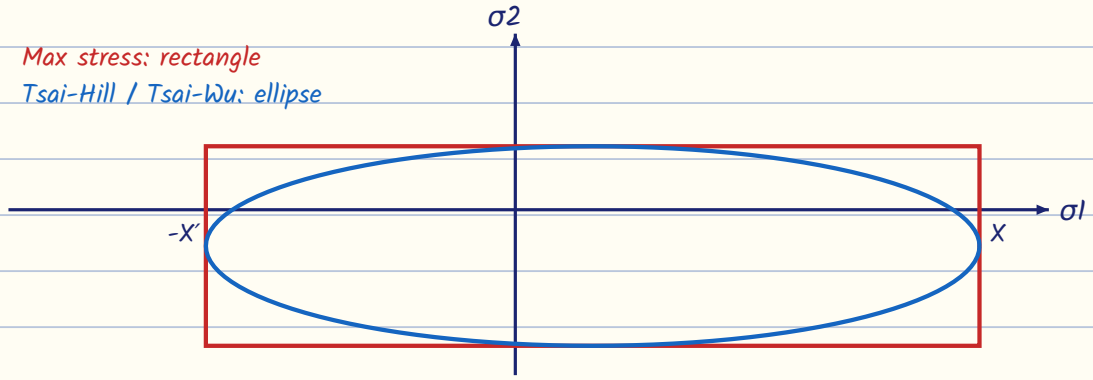


Fig. 3 Failure envelopes for 1-2 (12 = 0): rectangle (max stress) vs ellipse (interactive)

2.3 Hoffman criterion

Hoffman (1967) extended Hill's criterion with **linear terms** to differentiate between tensile and compressive strengths (like Tsai-Wu but with F_{12} fixed = $-\frac{1}{2}F_{11}$). Failure when the LHS = 1. For σ_1 = 500, σ_2 = 20, τ = 30 (glass/epoxy): 0.386 - 0.015 + 0.109 - 0.349 + 0.476 + 0.174 = **0.780 < 1** (safe).

III. Importance of Shear Strength

- The in-plane shear strength S is **small** (≈ 70 MPa for glass / epoxy, 70-100 MPa for carbon / epoxy) and is **matrix-dominated**; it is also non-linear, so shear controls failure for angles 10°-80°.
- **Shear stress** is the main load in ±45° laminates, torsion shafts, thin-wall tubes, joints and webs of beams. In a lamina at θ: $\tau_{12} = -\sigma_x \sin\theta \cos\theta$ - max at 45°.
- At θ = 45° the tension strength is only **57 MPa** (Tsai-Hill) - 5 % of X_t; failure is by shear and transverse tension in the matrix.
- **Interlaminar shear (ILSS)** between plies governs delamination, bolt bearing, impact damage and short-beam failure; typical ILSS of CFRP is 60-90 MPa.

- Shear strength is needed for each failure theory (S appears in all: S^2, $F_{66} = 1/S^2$) so it **must be measured accurately**.

Shear test	Principle	Remark
±45° tension test (ASTM D3518)	Tension on $[\pm 45]_s$ coupon: $S = P / 2A$	Simple; gives in-plane shear curve; large strain
Iosipescu V-notch (ASTM D5379)	Fixture loads a V-notched beam in pure shear	Uniform shear in test section; accurate G_{12}, S
Rail shear (2 or 3 rail)	Plate shear between steel rails	Good, but large specimen
Torsion of thin tube	Pure shear in a cylindrical wall	Best state of stress; expensive
Short-beam shear (ASTM D2344)	3-point bend with small span/depth ratio → interlaminar shear	Quality control of ILSS

IV. Choice of Failure Criteria

Criterion	Advantage	Disadvantage	Typical use
Max stress	Simple; gives failure mode; uses simple tests	No interaction; unconservative; kinks	Preliminary design, mode identification
Max strain	Includes Poisson effect; mode known	No interaction of stresses	Design with strain allowables (aerospace)
Tsai-Hill	Smooth; interaction; one equation	Equal tension and compression assumed; no mode	Ductile-matrix, UD laminae
Tsai-Wu	General; tension ≠ compression; invariant	F_{12} uncertain; no mode information	FEM codes, general laminate analysis
Hashin / Puck / LaRC	Mode-based with physical basis; better for compression	More inputs, complex	Advanced / progressive failure analysis

Guidelines for choosing

> Use **max stress / max strain** when failure **mode** is needed or for quick checks; use **Tsai-Wu / Tsai-Hill** for strength-ratio and envelopes.

> For brittle compression failure prefer criteria with different X_t, X_c (Tsai-Wu, Hoffman, Hashin); for ±θ laminates with equal strengths Tsai-Hill is adequate.

> Always combine with **ply-discount (progressive failure)** and **knock-down factors** (environment, fatigue, scatter, safety factor 1.5-2) for design.

> Compare with test results (World-Wide Failure Exercise: Puck, Tsai-Wu, Zinoviev, Cuntze perform best); **no single criterion is universally best**.

V. Examples

Example 1

Glass/epoxy (X_t = 1062, X_c = 610, Y_t = 31, Y_c = 118, S = 72 MPa). A lamina with fibres at 45° is loaded in uniaxial **compression** σ_x = -50 MPa. Check failure by Tsai-Hill and Tsai-Wu.

$\sigma_1 = \sigma_x \cos^2 45° = -25$, $\sigma_2 = \sigma_x \sin^2 45° = -25$, $\tau_{12} = -\sigma_x \sin 45° \cos 45° = +25$ MPa.

Tsai-Hill (compressive strengths X_c, Y_c): $(25/610)^2 - (25 \times 25)/610^2 + (25/118)^2 + (25/72)^2 = 0.0017 - 0.0017 + 0.0449 + 0.1206 = 0.165 < 1$ → safe; $R = 1/\sqrt{0.165} = 2.46$.

Tsai-Wu: $a = F_{ij}\sigma_i\sigma_j = 0.2795$, $b = F_i\sigma_i = -0.577$ → $a + b = -0.298 < 1$ → safe; $R = (0.577 + \sqrt{(0.333 + 1.118)})/0.559 = 3.19$.

For 45° **tension** the Tsai-Wu strength is only **56 MPa**: compressive loading is much safer here because $Y_c \gg Y_t$.

Example 2

Compare the strength of a glass/epoxy lamina at θ = 10°, 30°, 45°, 60° under uniaxial tension.

θ	10°	30°	45°	60°	90°
Max stress (MPa)	421	124	62	41.3	31
Tsai-Hill (MPa)	367.7	99.2	56.9	40.1	31
Failure mode (max stress)	Shear	Transverse	Shear	Transverse	Transverse

Strength falls from **1062 MPa at 0°** to < 40 % at 10° - the importance of fibre alignment and of shear.

Example 3

A lamina carries σ_1 = 600, σ_2 = 15, τ_{12} = 50 MPa. Does Tsai-Hill predict failure?

$(600/1062)^2 - 600 \times 15/1062^2 + (15/31)^2 + (50/72)^2 = 0.3192 - 0.0080 + 0.2341 + 0.4823 =$ **1.028 > 1** → **fails** (shear + transverse interaction), although max stress says safe (all components below strength: 600 < 1062, 15 < 31, 50 < 72). → max stress is **unconservative** here.

VI. Fibre-Matrix Load Transfer and Failure of Short-Fibre Lamina

When a fibre breaks (or ends), the load is **re-introduced** into it by interfacial shear stress τ_i. The length needed for the fibre stress to reach σ_{fu} is the **critical length** l_c; fibres shorter than l_c **pull out** instead of breaking.

$$l_c = \sigma_{fu}\, d / (2\,\tau_i) \quad \text{(force balance: } \sigma_{fu}(\pi d^2/4) = \tau_i(\pi\, d)(l_c/2))$$

$$\sigma_{lu} = \sigma_{fu} V_f (1 - l_c/2l) + \sigma'_m V_m \;\; (l > l_c) \qquad \sigma_{lu} = \tau_i (l/d) V_f + \sigma'_m V_m \;\; (l < l_c)$$

Example

- $d = 10\ \mu m$, $\sigma_{fu} = 3500$ MPa, $\tau_i = 50$ MPa → $l_c = 3500 \times 10/(2 \times 50) =$ **350 µm** ($l_c/d = 35$).
- For $l = 2$ mm: efficiency $= 1 - 350/(2 \times 2000) =$ **0.91** - short fibres with $l > 10\, l_c$ are almost as effective as continuous fibres.

6.1 Toughening mechanisms (energy absorption)

- **Fibre fracture** and **fibre pull-out** (work $\propto \tau_i\, l_c^2/d$): the main source of toughness in brittle-matrix composites.
- **Debonding** of the interface and **matrix cracking**, crack deflection / bridging by fibres.
- A **weak** interface improves toughness (crack deflection, pull-out) but reduces transverse and shear strength; a **strong** interface raises Y_t, S but gives brittle failure → compromise via fibre sizing / coupling agents.

VII. Failure of Laminates

- **First-ply failure (FPF)**: usually transverse cracking in the off-axis / 90° plies (Y_t, S exceeded).
- **Progressive failure**: failed plies are degraded (ply-discount), then loads are redistributed until the **last-ply failure (LPF)** by fibre fracture.
- **Delamination**: separation of plies by interlaminar stresses σ_z, τ_{xz}, τ_{yz}, which are large at **free edges**, holes, notches and under impact. Resisted by interlaminar toughness G_{Ic}, G_{IIc} (DCB and ENF tests).
- **Fatigue**: matrix cracks → delamination → fibre breaks (stiffness degrades gradually, no sudden failure as in metals); **impact (BVID)**: barely visible internal damage cuts compression strength by 30-50 % (CAI).

- **Environment**: moisture absorption and temperature lower matrix-dominated strengths (Y, S, X_c) - design with "hot-wet" allowables.

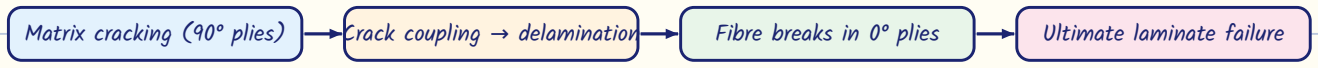


Fig. 4 Sequence of damage in a tensile cross-ply laminate

Damage mode	Cause	Remedy
Matrix cracking	Low Y_t, thermal stress	Toughened resins, ply thickness / angle design
Delamination	Interlaminar stress, impact	Stitching, z-pinning, interleaves, avoid free-edge
Fibre microbuckling	Compression with misalignment	Better fibre alignment, higher G_m, 0° ply support
Fibre breakage	Over-tension	Higher V_f, higher-strength fibre

VIII. More Solved Examples

Example 4

Glass/epoxy lamina with fibres at 30° is loaded by σ_x = 60 MPa tension. Find strength ratio by max stress, Tsai-Hill and Tsai-Wu.

$\sigma_1 = 0.75 \times 60 = 45$, $\sigma_2 = 0.25 \times 60 = 15$, $\tau_{12} = -0.433 \times 60 = -25.98$ MPa.

Max stress: $R = \min(1062/45, 31/15, 72/25.98) = \min(23.6, 2.07, 2.77) =$ **2.07** (matrix transverse tension).

Tsai-Hill: $(45/1062)^2 - (45 \times 15)/1062^2 + (15/31)^2 + (25.98/72)^2 = 0.0018 - 0.0006 + 0.2341 + 0.1302 = 0.3655 \rightarrow R =$ **1.65**.

Tsai-Wu: $a = 0.181$, $b = 0.325$, $a + b = 0.506 \rightarrow R = [-0.325 + \sqrt{(0.1056 + 0.7239)}]/0.362 =$ **1.62**. Failure stress: $\sigma_x = 60 \times R =$ **124 / 99 / 97 MPa** (max stress / Tsai-Hill / Tsai-Wu).

Example 5

Carbon/epoxy with E_1 = 181 GPa, X_t = 1500 MPa: find the ultimate strain along fibres and the stress in a 0° lamina at 0.5 % strain; will it fail?

$\varepsilon_{1t} = X_t/E_1 = 1500/181000 =$ **0.83 %**. At ε_1 = 0.5 %, $\sigma_1 = E_1\varepsilon_1 = 181 \times 0.005 =$ **905 MPa < 1500** → safe, $R = 0.83/0.5 =$ **1.66**.

Example 6

For σ_1 = 500, σ_2 = 20, τ_{12} = 30 MPa (glass/epoxy) apply Hashin matrix-tension criterion.

$(\sigma_2/Y_t)^2 + (\tau_{12}/S)^2 = (20/31)^2 + (30/72)^2 = 0.416 + 0.174 = \mathbf{0.590 < 1}$ → no matrix failure; fibre tension: $(500/1062)^2 + (30/72)^2 = 0.222 + 0.174 = 0.396 < 1$ ✓.

IX. Derivations Worth Knowing

9.1 Strength of off-axis lamina from Tsai-Hill

Put $\sigma_1 = \sigma_x m^2$, $\sigma_2 = \sigma_x n^2$, $\tau_{12} = -\sigma_x mn$ ($m = \cos\theta$, $n = \sin\theta$) in the Tsai-Hill equation:

$$\sigma_x^2[\, m^4/X^2 - m^2n^2/X^2 + n^4/Y^2 + m^2n^2/S^2 \,] = 1$$

$$\mathbf{1/\sigma_x^2 = m^4/X^2 + (1/S^2 - 1/X^2)\, m^2n^2 + n^4/Y^2}$$

At $\theta = 0$: $\sigma_x = X$; at $\theta = 90°$: $\sigma_x = Y$; in between the shear term $(1/S^2)$ makes the curve fall very steeply.

9.2 Tsai-Wu strength ratio

Scale the applied stresses by R ($\sigma \to R\sigma$) and put into the Tsai-Wu equation: $\mathbf{aR^2 + bR - 1 = 0}$, with $a = F_{ij}\sigma_i\sigma_j$ and $b = F_i\sigma_i$. Positive root: $R = [-b + \sqrt{(b^2 + 4a)}] \,/\, 2a$; if $a = 0 \to R = 1/b$. Failure stresses = R × applied stresses.

9.3 Strain magnification factor (transverse loading)

Under σ_2 the lamina strain $\varepsilon_2 = \sigma_2/E_2$ is shared: $\varepsilon_2 h = \varepsilon_f V_f h + \varepsilon_m V_m h$ with $\varepsilon_f = \varepsilon_m E_m/E_f$. Hence

$$\varepsilon_m / \varepsilon_2 = 1 / [1 - V_f(1 - E_m/E_f)]$$

For glass/epoxy $V_f = 0.6$: $S_m = 1/[1 - 0.6(1 - 3.5/72)] = \mathbf{2.3}$; with epoxy fail-strain 4 % → $\varepsilon_{2u} \approx$ 1.7 % at best, and in practice 0.4-0.5 % because of voids and interface flaws.

9.4 Failure envelope in 1-12 plane

For $\sigma_2 = 0$, Tsai-Hill reduces to $(\sigma_1/X)^2 + (\tau/S)^2 = 1$: an **ellipse** with semi-axes X and S; max stress gives a **rectangle**. Interaction shrinks the safe area at the corners.

Quantity	Glass/epoxy	Carbon/epoxy (T300)	Kevlar/epoxy
X_t (MPa)	1062	1500	1400
X_c (MPa)	610	1500	235
Y_t (MPa)	31	40	12
Y_c (MPa)	118	246	53
S (MPa)	72	68	34
X_t/Y_t	34	38	117

Observe: Aramid has very low X_c (kink bands in fibre) - never use Kevlar where compression governs.

X. Design Practice Against Failure

- Orient plies in **several directions** ($[0/\pm45/90]_s$) so that no ply sees large Y or S stresses.
- Keep at least **10 % of plies in each of 0°, ±45°, 90°** (rule of thumb) for damage tolerance.
- Avoid **free-edge** stacks with many same-angle plies together (< 4 adjacent plies); use ply drop-off tapers, stitching or edge caps.
- Use **strain allowables** (0.3-0.4 % for damage-tolerant CFRP) rather than full ultimate strain; apply **safety factor 1.5** on ultimate.
- Test: coupons → elements → sub-components → full scale (**building-block approach**).

XI. Quick Revision - Unit V

Remember

> X_t: fibre fracture; X_c: microbuckling ($G_m/(1 - V_f)$); Y_t, S: matrix / interface; Y_c: matrix shear at 45°.

> Y_t and S are tiny relative to X_t → **first failure is matrix cracking**; angle misalignment of 5-10° drastically lowers strength.

> Max stress & strain (non-interactive, mode known); Tsai-Hill, Hoffman, Tsai-Wu (interactive); Hashin / Puck (mode-based).

> Shear strength tests: ±45° tension, Iosipescu, rail shear, short-beam shear.

Likely exam questions

> Explain micromechanics of failure of a unidirectional lamina in tension, compression and shear.

> Discuss anisotropic strength and failure theories; compare Tsai-Hill, Hoffman and Tsai-Wu.

> Why is shear strength important? How is it measured?

> How do you choose a failure criterion? Solve examples.

Short Answer Questions (2 marks)

Q1. What is the main failure mode in longitudinal compression?

Ans: Microbuckling (kinking) of fibres.

Q2. Why is Y_t so low?

Ans: Matrix strain magnification, interface debonding and voids.

Q3. Write Rosen's shear-mode buckling stress.

Ans: $\sigma_c = G_m/(1 - V_f)$.

Q4. Define critical fibre length.

Ans: $l_c = \sigma_{fu} d/(2\tau_i)$: the minimum length for fibre stress to reach its strength.

Q5. How does misalignment affect strength?

Ans: Introduces shear and transverse stresses, lowering strength sharply (≈ 60 % at 10°).

Q6. Name two shear test methods.

Ans: ±45° tension test, Iosipescu V-notch test (also rail shear).

Q7. Why is Tsai-Wu preferred in FEM?

Ans: It is a single smooth invariant criterion with tension-compression differences and interaction.

Q8. Which criterion gives mode of failure?

Ans: Maximum stress / strain and Hashin / Puck criteria.

Q9. What is ply discount?

Ans: Reducing the stiffness of failed plies in progressive failure analysis.

Glossary of Key Terms

Term	Meaning
Microbuckling	Local instability of fibres in compression, forming kink bands
Ineffective length	Length of broken fibre over which load is not fully carried
Strength ratio R	Failure load / applied load (R > 1 safe)
Free-edge effect	High interlaminar stresses near laminate edges causing delamination
ILSS	Interlaminar shear strength from short-beam test
CAI	Compression after impact strength
BVID	Barely visible impact damage
Progressive failure	Sequential ply failure with stiffness degradation

Answer Outlines for 10-mark Questions

Micromechanics of failure of a UD lamina

> Sketch of 5 failure modes.

> Tension: fibre strength, V_{min}, V_{crit}; compression: Rosen modes; transverse tension / compression; shear.

Compare failure theories

> Max stress, max strain, Tsai-Hill, Hoffman, Tsai-Wu equations.

> Envelope sketch, interaction, tension ≠ compression, mode information.

Importance of shear strength

- Low S, matrix dominated; off-axis loading; torsion, ±45° laminates.
- Tests: ±45°, Iosipescu, rail shear; ILSS.